O-RAN.WG6.TS.O-Cloud-IM.0-R004-v05.00

Technical Specification

O-RAN Work Group 6 (Cloudification and Orchestration)

O-Cloud Information Model

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Foreword

This Technical Specification (TS) has been produced by WG6 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

The O-Cloud Information Model provides the logical model of information elements and their relationships. This includes:

- Enumerations which provide a predefined list of choices.
- Information Object Classes intended to convey classes that can become an Information Object
- Data Types which provide structure to the elements of an Information Object Class
- Notifications which is the information conveyed in a message from a service publisher to a service consumer

Introduction

The O2 interface exposes information about the O-Cloud to external clients, presumably the SMO as the primary consumer. There are many relationships between elements of the model which need to be captured.

The O-Cloud Information Model is based on the 3GPP process defined to document a Network Resouce Model (NRM). The content contains all the salient elements of the NRM but does not follow the strict outline defined in 3GPP TS 32.160 [i.1]. The outline has been modified to align with other WG6 documents and to eliminate duplicative maintenance which leads to discord within the model.

UML provides a rich set of concepts, notations, and model elements to model distributive systems. 3GPP TS 32.156 [1] provides the necessary and sufficient set of UML notations and model elements, including the ones built by the UML extension mechanism <<stereotype>> to model network management systems and their managed nodes. These conventions are applied to the O-Cloud Information Model.

The Information model uses embedded PlantUML for diagraming. The information model relationships are modeled using UML Class diagram conventions.

1 Scope

The contents of the present document are subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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This document is both a Specification and an Informational Report in that it specifies the Information Model that is foundational for O-RAN's model-driven architecture for the O-Cloud information exposed over the O2 interface.

In addition, this document includes information about existing standards and industry work that serve as a basis for work items in O-RAN.

Lastly, this document describes the de facto methods and procedures with respect to a "modeling continuum" that aims to establish and evolve an O-Cloud Information Model from which O-RAN Data Models may be generated manually or with a set of tools.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] 3GPP TS 32.156: "Fixed Mobile Convergence repertoire ".

[2] O-RAN.WG10.TS. Information Model and Data Models: "O-RAN Information Model and Data Models Specification".

[3] O-RAN.WG6.TS.Orch-Use-Cases: "Cloudification and Orchestration Use Cases and Requirements for O-RAN Virtualized RAN"

[4] OMG: "Unified Modelling Language Specification, Version 2.5.1". https://www.omg.org/spec/UML/2.5.1.

[5] IETF RFC 5424 : "The Syslog Protocol"

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 32.160: " Management service template".

[i.2] O-RAN.WG6.TS.O2-GA&P: “O2 Interface General Aspects and Principles”.

[i.3] O-RAN.WG6.TS.CADS: "Cloud Architecture and Deployment Scenarios".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

Generalized Object This is a kind of object whereby specialized objects can be derived from.

Specialized Object This is a kind of object which is derived from one or more Generalized Object(s).

3.2 Symbols

For the purposes of the present document, the following symbols apply:

3.3 Abbreviations

For the purposes of the present document, the [following] abbreviations [given in ... and the following] apply:

AAL Accelerator Abstraction Layer
OIM O-Cloud Information Model

4 O-Cloud Information Model

4.1 Information Modeling Conventions

4.1.1 Modeling approach, Unified Modeling Language (UML)

The Information Model within O-Cloud Infrastructure shall use the Unified Modeling Language™ (UML®) version 2.5.1 [4] from the Object Management Group (OMG).

The O-Cloud Infrastructure Information Model shall follow methodology documented in 3GPP TS 32.160 [i.1] Clause 5.2.

4.1.2 Namespaces in the Information Model

A Namespace provides a container for named elements, which are called its members. A Namespace may also import named elements from other specifications, in which case those elements become members of the importing Namespace.

4.1.3 Namespace Relationships

Figure 4.1.3-1 describes the relationship between each namespace defined in this specification.

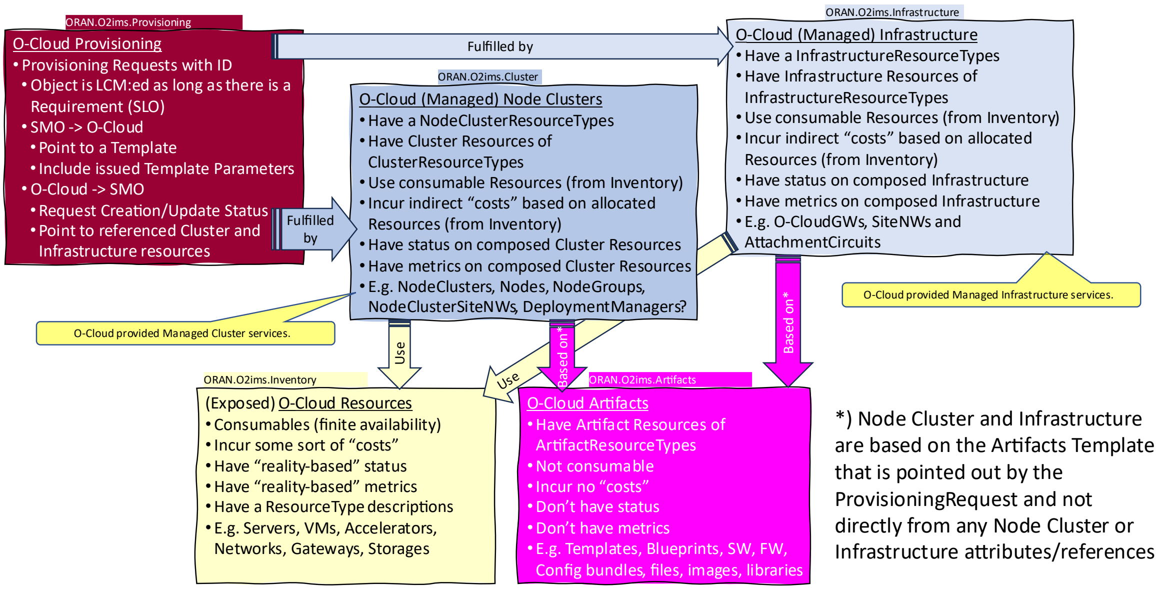

Figure 4.1.3-1 Relationship between Namespaces

4.2 Information Model Definitions

4.2.1 Namespace: ORAN.O2ims.Inventory

4.2.1.1 Namespace Overview

The O-RAN O2ims Inventory namespace defines the classes and their attributes which are directly exposed to the consumer (SMO) via its O2 IMS Interface.

This namespace is used for O-Cloud infrastructure Resources. That content is meant only to be internally consumable to the O-Cloud. The O2ims Inventory has status, metrics and types as well as its topologies. These resources can be used to build up the O-Cloud Node Clusters and their compute, networking and storage capabilities.

4.2.1.2 Imported and associated information entities

4.2.1.2.1 Imported information entities and local labels

Information entities imported into the model are also elements of the namespace and therefore shall remain unique within the namespace.

Table 4.2.1.2.1-1 Imported Entities

Label reference	Local label
TS 28.622 [6], information object class, `Top`	`Top`

4.2.1.2.2 Associated information entities and local labels

Information entities referenced in the model through an association which may be extended and overloaded with a new entity of the same name within the namespace are listed in the associated entities table.

Table 4.2.1.2.2-1 Associated Entities

Label reference	Local label
WG10 IM/DM [2], PerformanceMeasureDictionary	PerformanceMeasureDictionary
WG10 IM/DM [2], *PerformanceMeasureDefinition_*	PerformanceMeasureDefinition
WG10 IM/DM [2], AlarmDictionary	AlarmDictionary
WG10 IM/DM [2], AlarmDefinition	AlarmDefinition
WG10 IM/DM [2], ProbableCause	ProbableCause
4.2.2.4.1, ArtifactResourceType	ArtifactResourceType
4.2.2.4.1, ArtifactResource	ArtifactResource
4.2.3.4.1, ClusterResourceType	ClusterResourceType
4.2.3.4.2, ClusterResource	ClusterResource
4.2.3.4.3, NodeClusterType	NodeClusterType
4.2.3.4.4, NodeCluster	NodeCluster
4.2.4.4.1, InfrastructureResourceType	InfrastructureResourceType
4.2.4.4.2, InfrastructureResource	InfrastructureResource

4.2.1.3 Class Diagram

4.2.1.3.1 Relationships

4.2.1.3.1.1 Description

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for the O-RAN Cloud (O-Cloud). This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes. The UML Class diagram depicts a view of the namespace maintaining a defined focus so as not to overcrowd a single diagram trying to depict all relationships within the namespace.

4.2.1.3.1.2 O2ims Inventory Object Relationships

The O2ims services have operations on the following managed service objects that are provided by the O-Cloud:

- OCloud,
- Location,
- OCloudSite,
- ResourceType,
- ResourcePool,
- Resource,
- DeploymentManager, and
- InventoryChangeSubscription

Figure 4.2.1.3.1.2-1 illustrates the relationships in between the information entities of the O2ims_InfrastructureInventory services. Figure 4.2.1.3.1.2-2 illustrates the relationship for the subscribe/notify pattern.

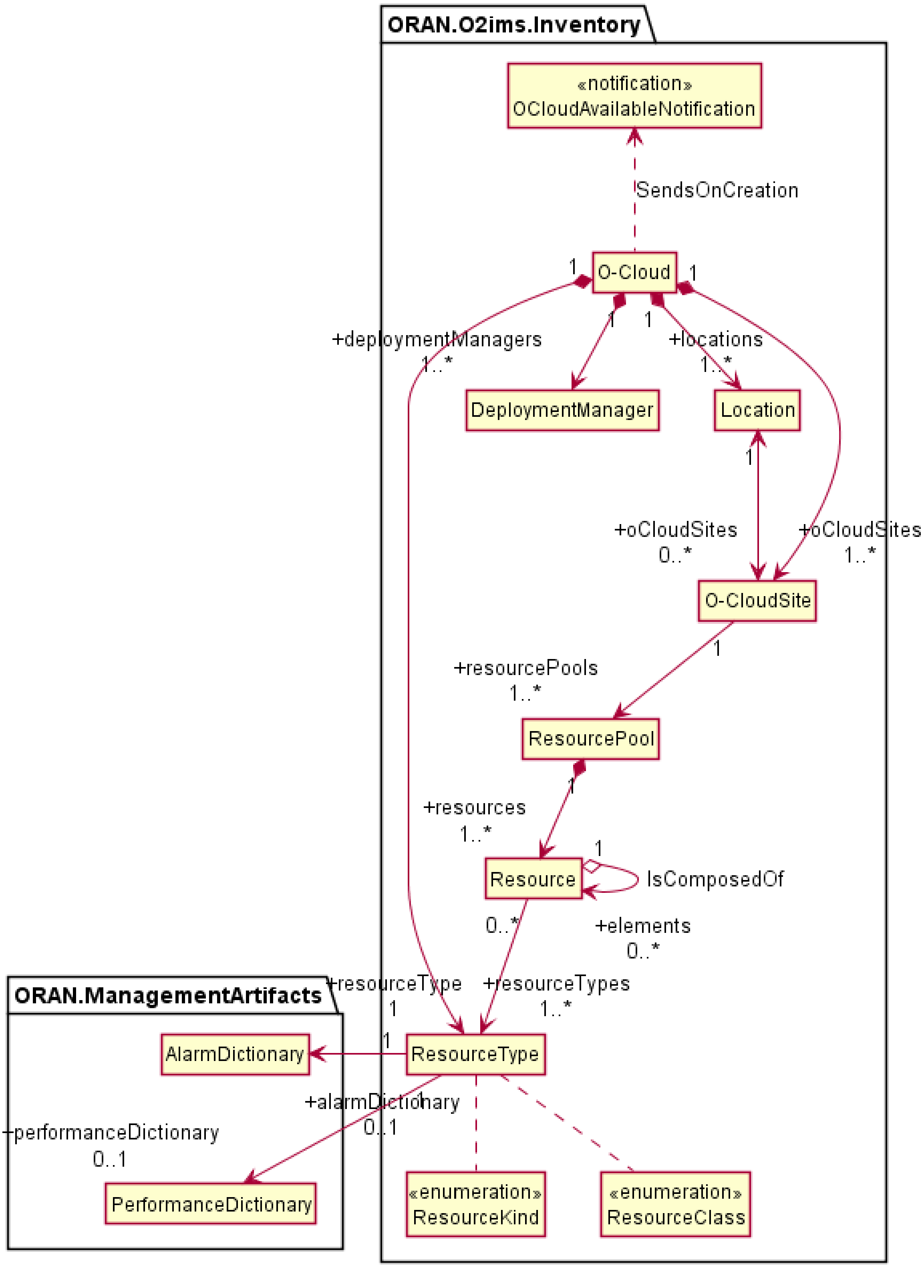

Figure 4.2.1.3.1.2-1 O2ims Inventory Objects Relationship Views

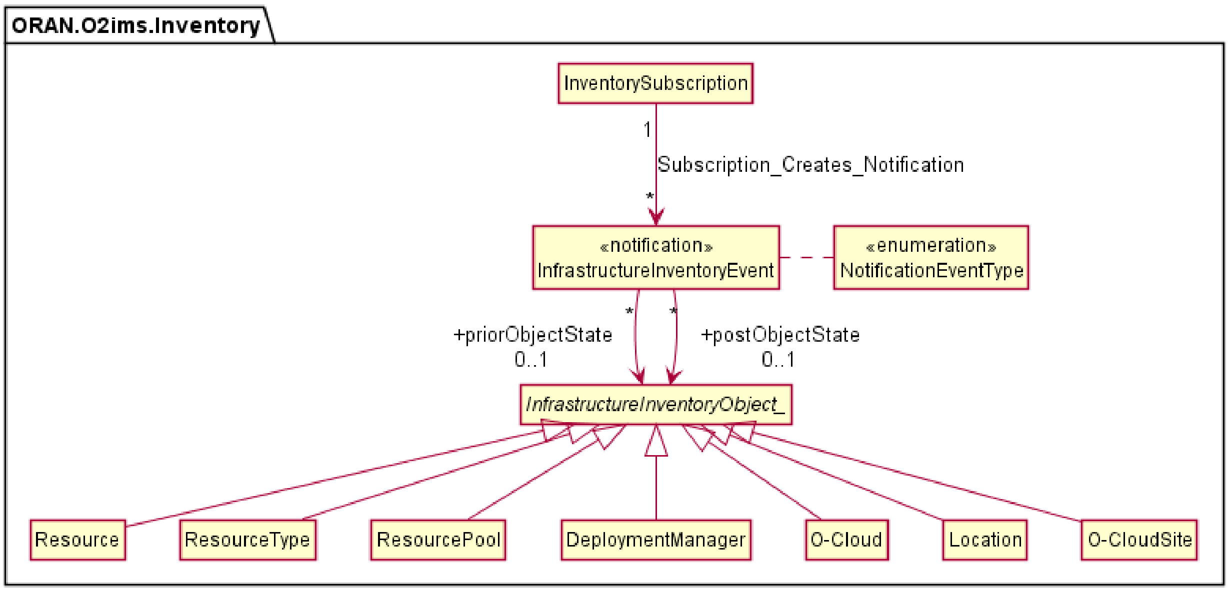

Figure 4.2.1.3.1.2-2 O2ims Inventory Event Subscribe/Notify Relationship View

4.2.1.3.1.3 Infrastructure Monitoring Relationships

4.2.1.3.1.3.1 Infrastructure Fault Relationships

The O2ims_InfrastructureMonitoring services has operations on the following managed service objects that are provided by the O-Cloud for faults:

- AlarmList
- AlarmSubscription
- LogQuery

Figure 4.2.1.3.1.3.1 -1 illustrates the relationships in between the information entities of the O2ims_InfrastructureMonitoring fault services.

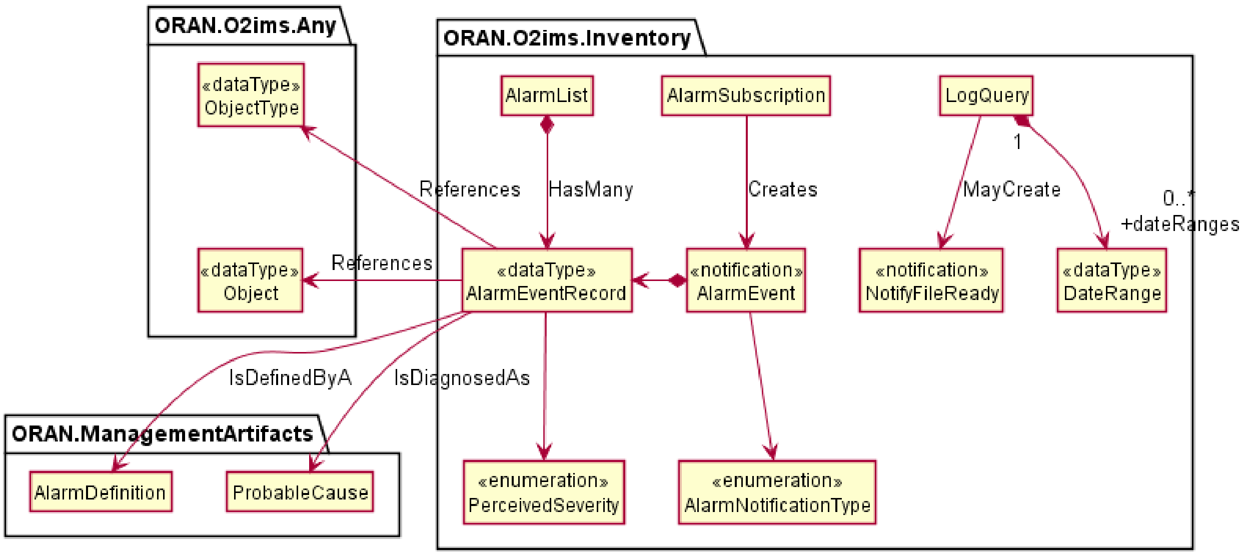

Figure 4.2.1.3.1.3.1-1 O2ims Alarm Relationship View

4.2.1.3.1.3.2 Infrastructure Performance Relationships

The O2ims_InfrastructureMonitoring services follow a flow described in the O2 GAP[i.2] clause 3.9.9. This primarily consists of:

1) The Performance Measurement Store - used for storage and configuration of performance measurements.

2) The Performance Measurement Job - used to collect performance measurements and insert them into the Performance Measurement Store.

3) The Performance Subscription - used to identify what PerformanceMeasurementRecord(s) in the PeformanceMeasurementStore are reported, when they are reported, and how the report is to be delivered to the subscriber.

4) The Performance Measurement Report - The notification payload used to convey measurement values identified by a Performance Subscription to the identified subscriber in a format selected by the subscriber.

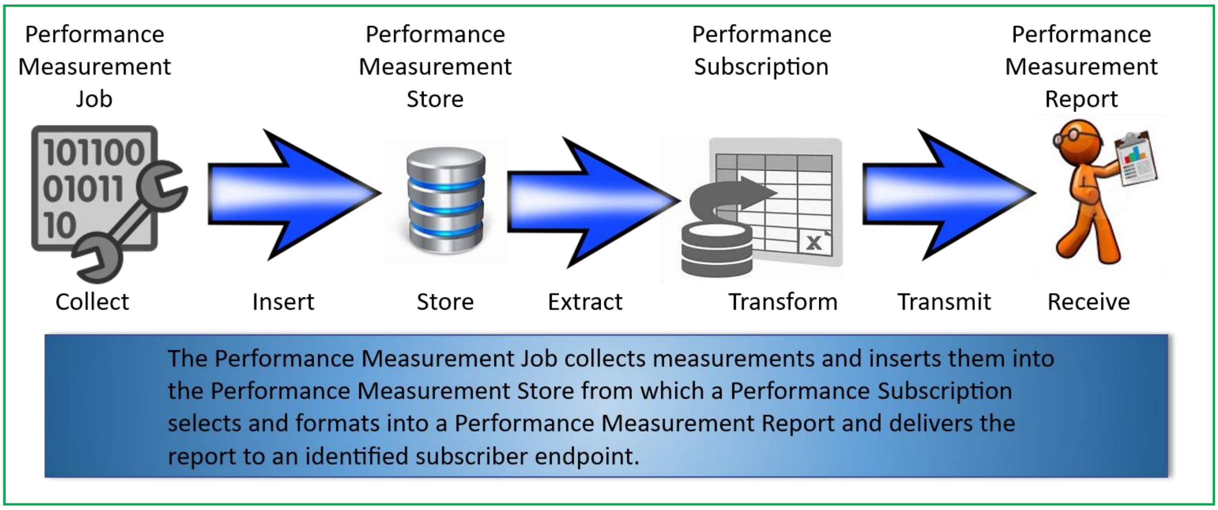

Figure 4.2.1.3.1.3.2-1 Performance Monitoring Information Flow

The O2ims_InfrastuctureMonitoring information model have sub-components as follows:

- Performance Store
  - PerformanceMeasurementStore
  - PerformanceMeasurementRecord
- Performance Jobs
  - PerformanceMeasurementJob
  - MeasuredResource
  - CollectedMeasurement
  - PerformanceMeasurementJobChangeSubscription
  - PerformanceMeasurementJobChangeNotification
- Performance Subscriptions
  - PerformanceSubscription
  - PerformanceReportingFrequency
  - PerformanceSubscriptionCriteria
- Performance Reports
  - PerformanceMeasurementReport
  - JobReport
  - ResourceReport
  - MeasurementValue
  - NotifyFileReady
  - FileInfo

Figure 4.2.1.3.1.3.2-2 illustrates the relationships in between the PerformanceMeasurementStore and PerformanceMeasurementRecord and other information entities.

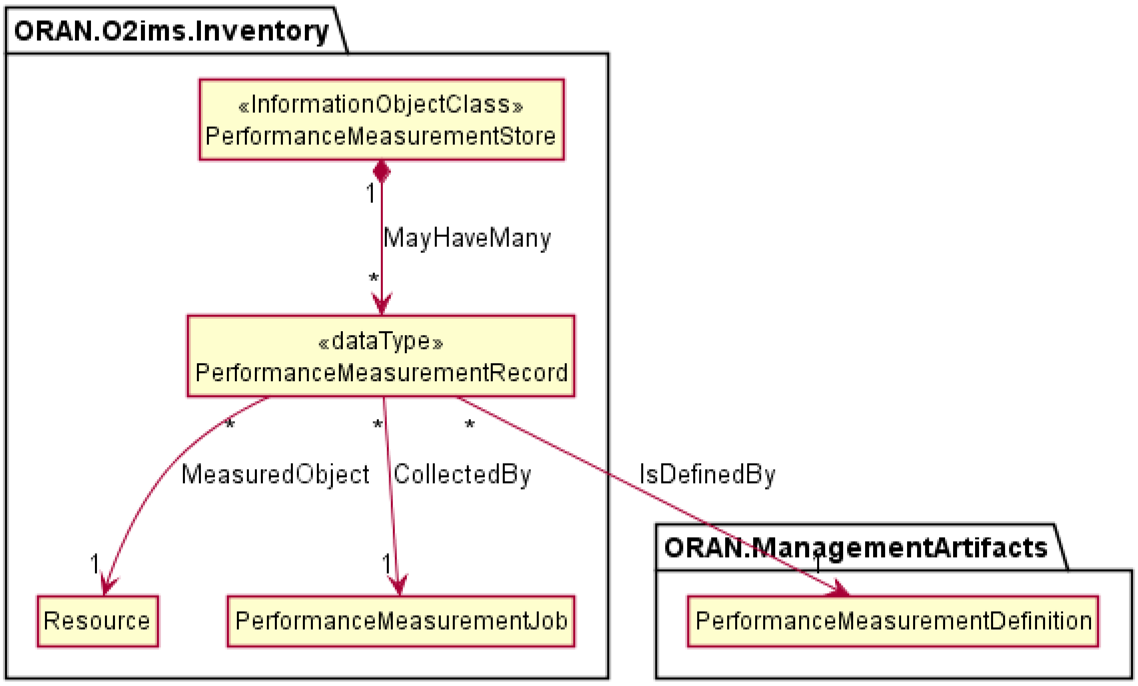

Figure 4.2.1.3.1.3.2-2 O2ims Performance Store Relationship View

Figure 4.2.1.3.1.3.2-3 illustrates the relationships in between the PerformanceMeasurementJob, MeasuredResource, CollectedMeasurement, PerformanceMeasurementJobChangeSubscription and other information entities.

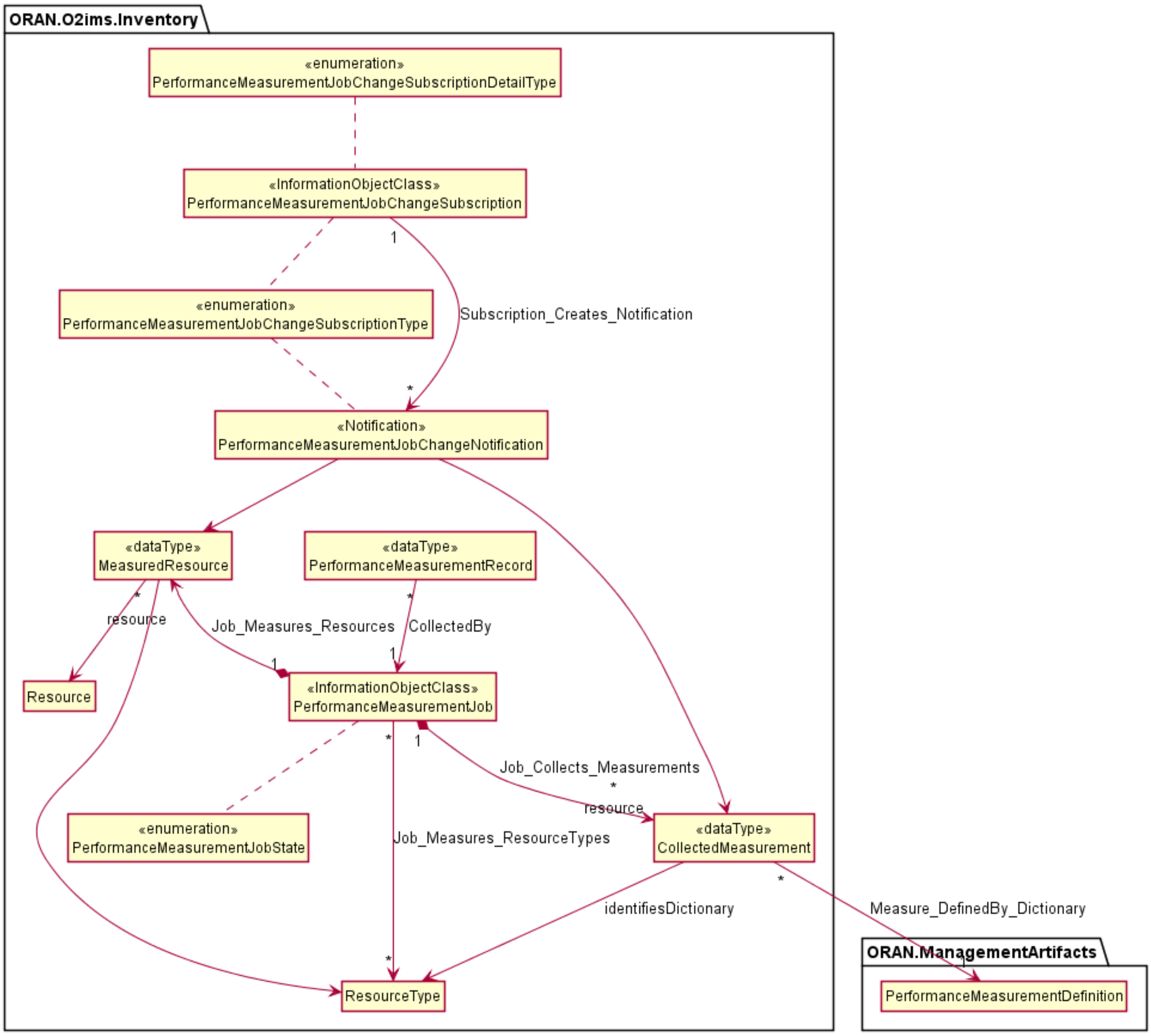

Figure 4.2.1.3.1.3.2-3 O2ims Performance Measurement Job Relationship View

Figure 4.2.1.3.1.3.2-4 illustrates the relationships in between the PerformanceSubscription, PerformanceReportingFrequency, PerformanceSubscriptionCriteria and other information entities.

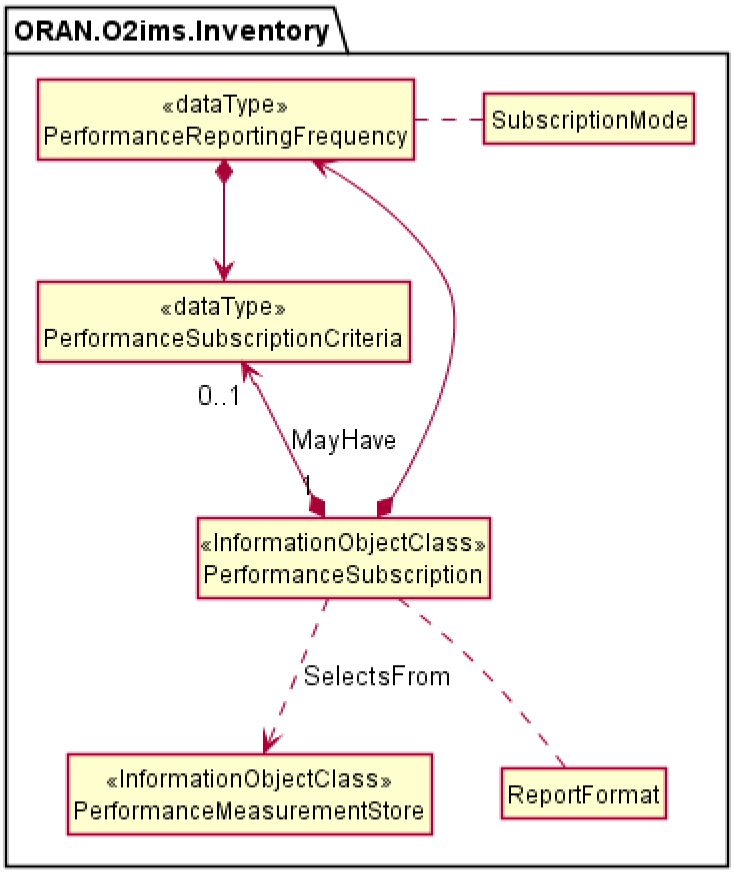

Figure 4.2.1.3.1.3.2-4 O2ims Performance Subscription Relationship View

Figure 4.2.1.3.1.3.2-5 illustrates the relationships in between the PerformanceMeasurementReport, JobReport, ResourceReport, MeasurementValue, NotifyFileReady, FileInfo and other information entities.

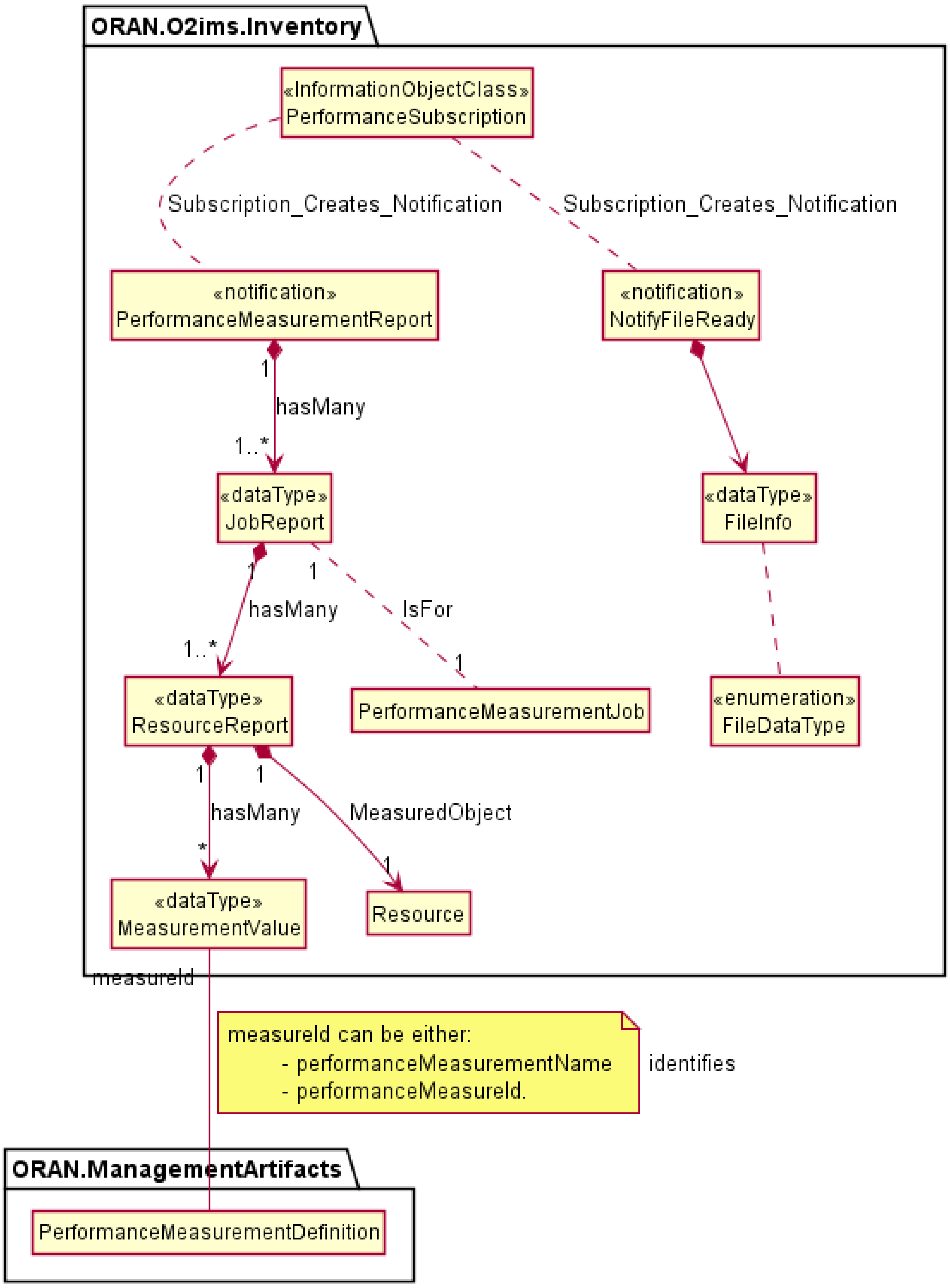

Figure 4.2.1.3.1.3.2-5 O2ims Performance Report Relationship View

4.2.1.3.1.4 O-Cloud Networking Relationships

4.2.1.3.1.4.1 O-Cloud Networking Relationships Overview

The O-Cloud Network provides seamless connectivity within an O-Cloud Site and facilitates external connectivity, such as transport network access, to various O-Cloud Resources. Clause 3.4.3 of the O-RAN.WG6.O2-GA&P [i.2] specification, outlines the fundamental concepts of the O-Cloud provisioning and defines these concepts at different layers. For instance, networking-related concepts such as O-Cloud Site Network Fabric (aka. SiteNWFabric) and O-Cloud Site Gateway (aka. SiteGW) can be defined at the hardware layer, which is related to that layers' consumable constructs, and the networking-related concepts such as O-Cloud Site Network (aka. SiteNW), O-Cloud Gateway, and O-Cloud AttachmentCircuit can be defined at the execution environment layer as described in the O-RAN.WG6.O2-GA&P [i.2] spec. Also, there can be inter-namespace relationships for instance, Networking concept related to an O-Cloud Node Cluster such as, O-Cloud Node Cluster Site Network (aka. NodeClusterSiteNW) defined in ORAN.O2ims.Clusters namespace mapped to a SiteNW object and terminated by an O-Cloud Gateway object defined in ORAN.O2ims.Inventory namespace.

An O-Cloud AttachmentCircuit as defined in O-RAN.WG6.O2-GA&P [i.2], clause 3.4.3.5.10 has two possible associations — one with an O-Cloud Gateway and another with the SiteNW. This accommodates the two scenarios where a SiteNW connects to the transport network through an O-Cloud Gateway or where a SiteNW connects directly to the transport network. There can also be the case that SiteNWs are only used for internal connectivity within an O-Cloud Site, in which case, there is no need to associate (neither directly, nor via an O-Cloud Gateway) a SiteNW to an O-Cloud AttachmentCircuit.

Figure 4.2.1.3.1.4-1 illustrates the O-Cloud networking-related Resource relationships between the Information Objects Classes.

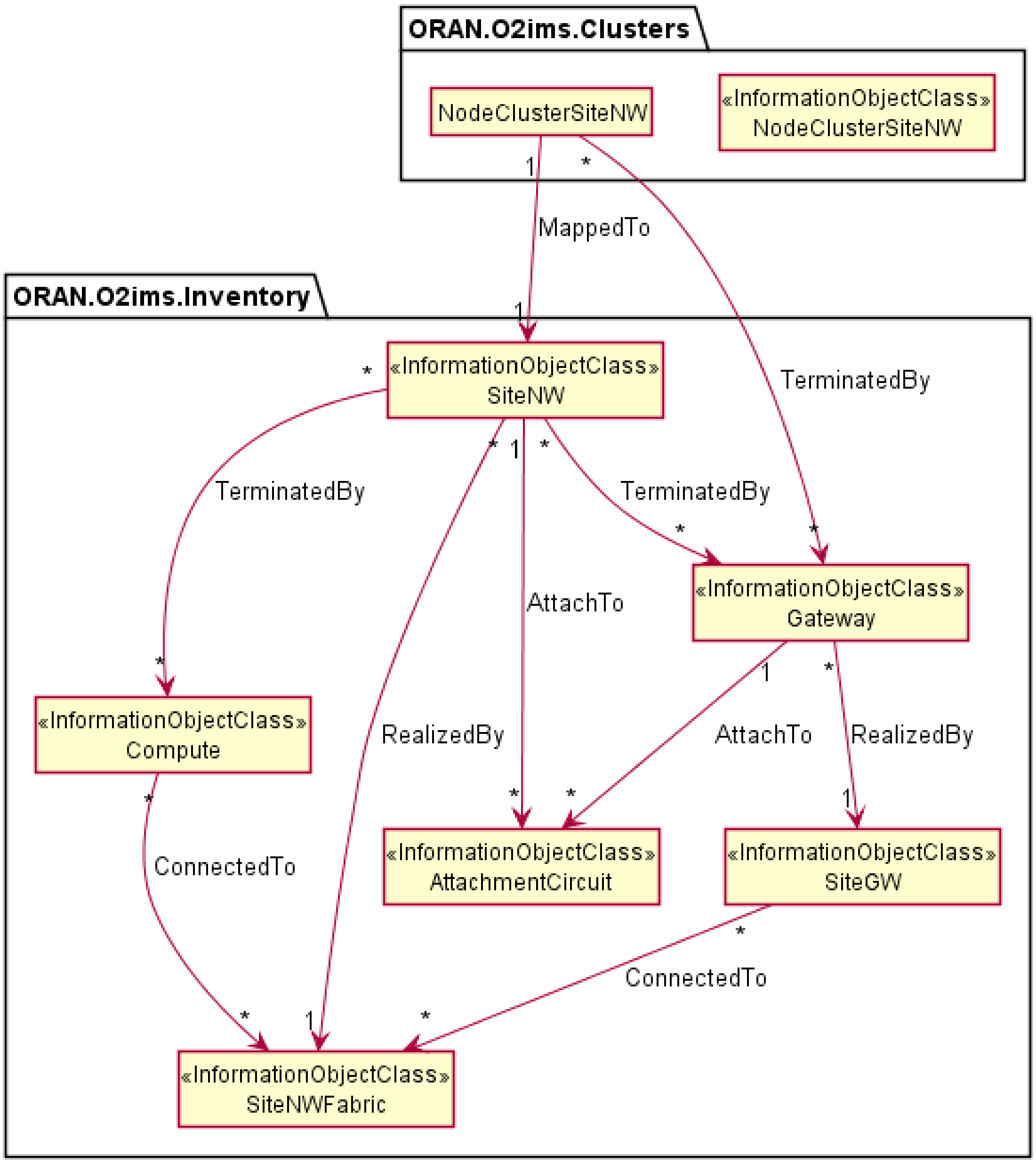

Figure 4.2.1.3.1.4.1-1 O-Cloud networking-related Resource relationships between the Information Object Classes.

4.2.1.3.1.4.2 O-Cloud Networking Relationships View

4.2.1.3.1.4.2.1 Overview

The O-RAN.WG6.O2-GA&P [i.2], clause 3.4.3.3 outlines fundamental concepts of the Resource View. For enabling separation of authority and operations, O-Cloud offers various types of resource views, in the form of different group of actors, some of them deals with hardware or more physical types of O-Cloud Resources whereas, others consume logical set of O-Cloud Resources.

4.2.1.3.1.4.2.2 Hardware Layer View

The O-RAN.WG6.O2-GA&P [i.2], clause 3.4.3.4 explains the Hardware Layer concept, which forms the foundation of a cloud, comprising physical computes that are inter-connected through network fabric and routers for external connectivity. According to O-RAN.WG6.CADS [i.3], a Cloud Site hosts Cloud Infrastructure equipment, abstracting O-Cloud Resources for leasing through the IMS. From the SMO perspective, understanding hardware concepts focuses on O-Cloud Resource locations, capabilities, capacity, availability, and potential disturbances, rather than physical realization.

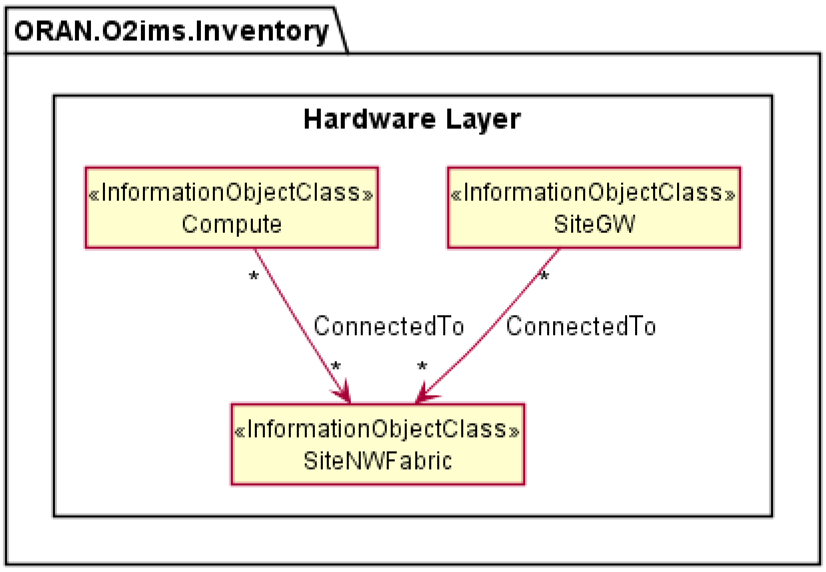

Figure 4.2.1.3.1.4.2.2-1 Hardware Layer View and networking related Information Object Classes

4.2.1.3.1.4.2.3 Execution Environment Layer View

The O-RAN.WG6.O2-GA&P [i.2], clause 3.4.3.5 explains the Execution Environment Layer concept, which transforms hardware layer constructs into consumable O-Cloud Node(s), Site Network(s), Gateway(s), and many more, as abstract O-Cloud Resources available for leasing through the IMS.

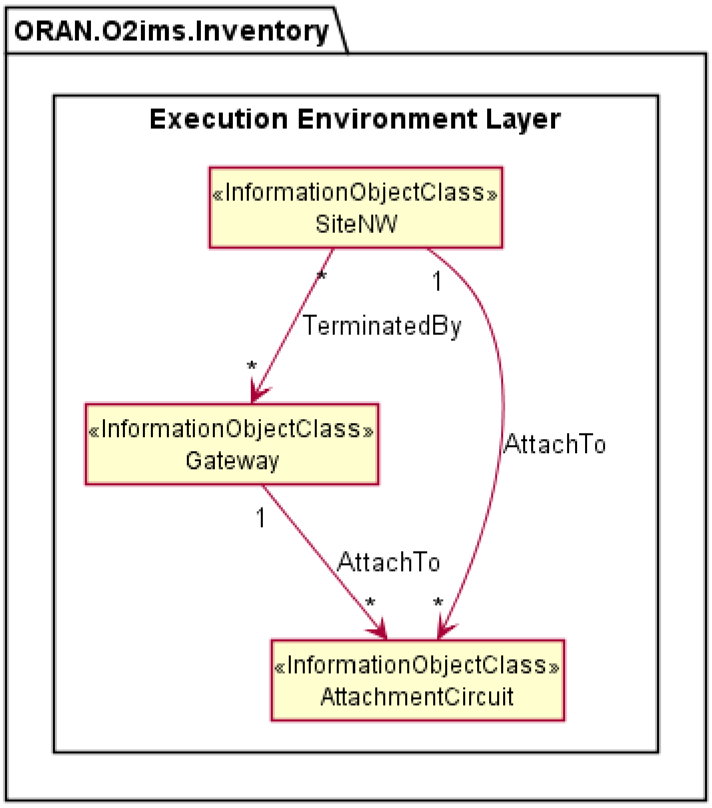

Figure 4.2.1.3.1.4.2.3-1 Execution Environment Layer View and networking related Information Object Classes

4.2.1.3.2 Inheritance

4.2.1.3.2.1 O2ims Inventory Inheritance

This subclause depicts the inheritance relationships of inventory object classes.

This subclause depicts the inheritance relationships.

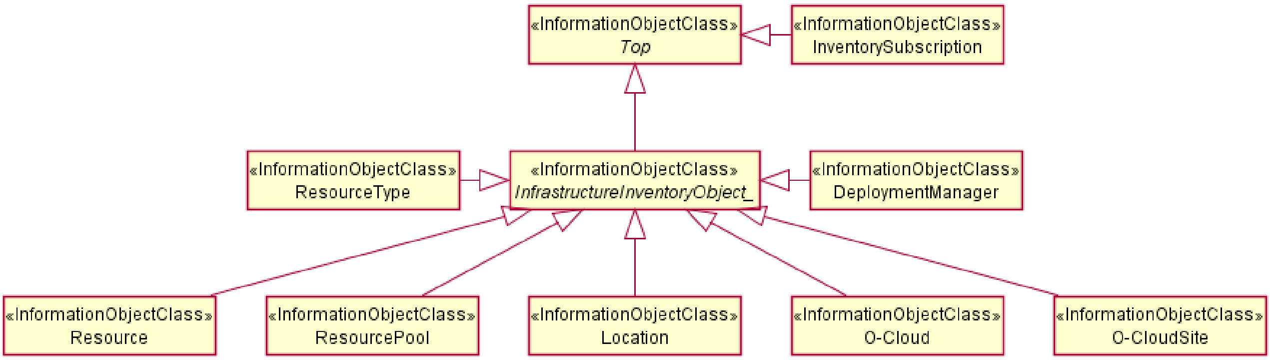

Figure 4.2.1.3.2.1-1 O2ims Inventory Intra-Namespace Inheritance

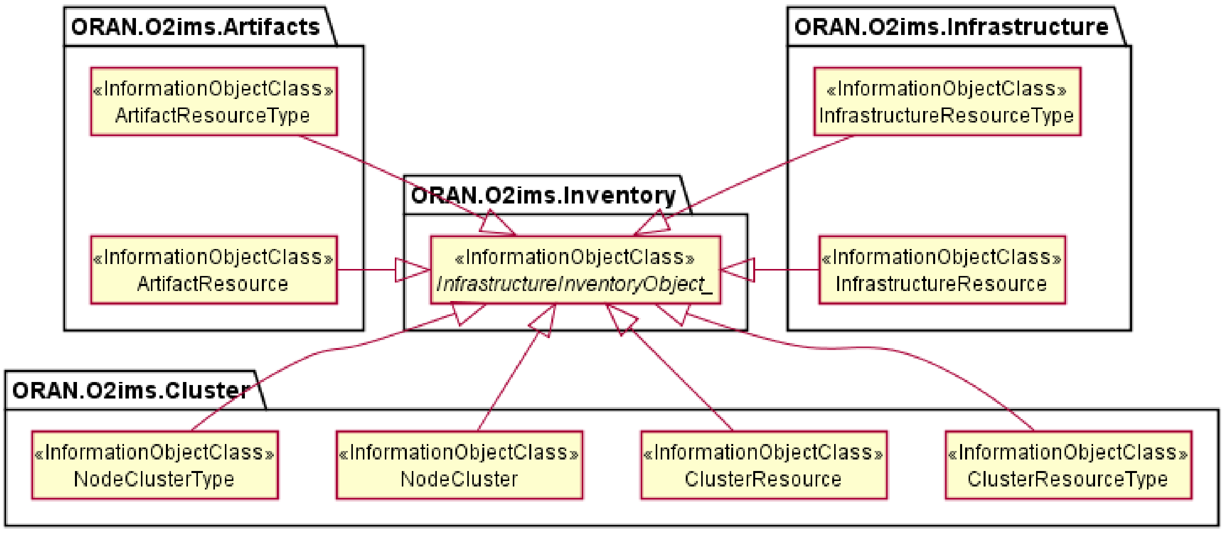

Figure 4.2.1.3.2.1-2 O2ims Inventory Inter-Namespace Inheritance

4.2.1.3.2.2 Infrastructure Monitoring Inheritance

This subclause depicts the inheritance relationships of alarm and performance.

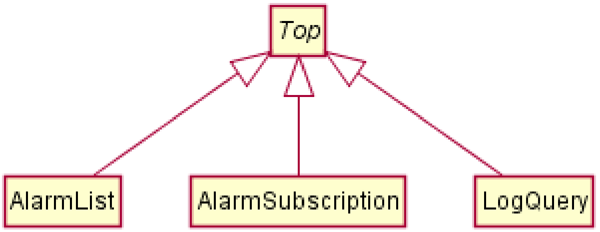

Figure 4.2.1.3.2.2-1 O2ims Alarm Inheritance

Figure 4.2.1.3.2.2-2 depicts the inheritance relationships of Performance.

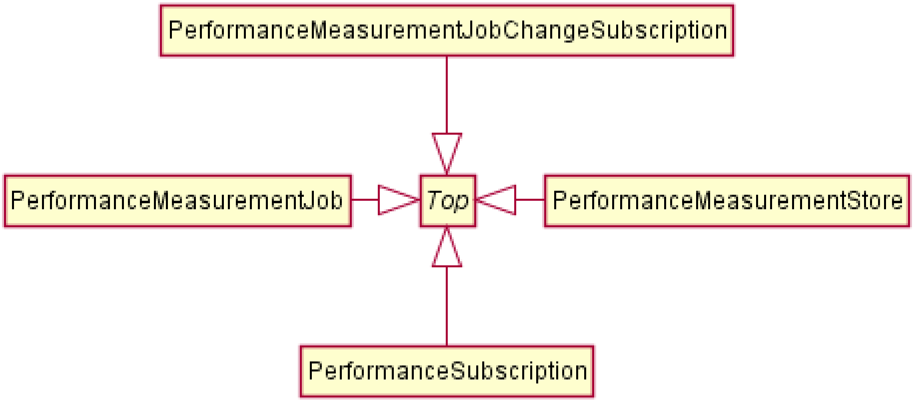

Figure 4.2.1.3.2.2-1 O2ims Performance Inheritance

4.2.1.4 Class Definitions

4.2.1.4.1 OCloud

4.2.1.4.1.1 Definition

The OCloud represents an instance of an O-Cloud. For more information about O-Cloud, refer to the "O-RAN O2 General Aspects and Principles Specification", clause 3.1 [i.2].

4.2.1.4.1.2 Attributes

Table 4.2.1.4.1.2-1 Attributes Properties for OCloud

Attribute name	Data Type/Description	Properties

oCloudId	**Data Type:** string
	
	Description: Identifier of the O-Cloud instance. Internally generated within an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

globalCloudId	**Data Type:** string
	
	Description: Identifier of the O-Cloud instance assigned by the SMO. This identifier is globally unique across O-Cloud instances known to the SMO. This value was provided by the SMO at cloud genesis and is stored in the O-Cloud IMS Inventory.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Human readable name of the O-Cloud as identified by the SMO at cloud genesis.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the O-Cloud as provided by the SMO at cloud genesis.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

infrastructureManagementServicesEndPoint	**Data Type:** string
	
	Description: The URI root to all services provided by the O2ims interface. Inventory is one of these services.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

resourceTypes	**Data Type:** array
	
	Description: List of Resource types.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items: ResourceType`

Attribute name	Data Type/Description	Properties
locations	**Data Type:** array **Description:** List of locations where O-Cloud Sites are deployed or can be deployed.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: Location
oCloudSites	**Data Type:** array **Description:** List of O-Cloud Sites that are known by the IMS and need to be exposed to the SMO for Homing.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: OCloudSite
deploymentManagers	**Data Type:** array **Description:** List of Deployment managers.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: DeploymentManager
smoRegistrationService	**Data Type:** string **Description:** IP address endpoint or the URL of the SMO.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uri
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the O-Cloud.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.1.3 Attribute constraints

None.

4.2.1.4.1.4 State diagram

None.

4.2.1.4.2 ResourceType

4.2.1.4.2.1 Definition

The ResourceType represents a resource type in an O-Cloud.

Resource types provide the categorization of O-Cloud Nodes as described in the O-RAN Cloud Architecture and Deployment Scenarios [i.3] specification. The values for these types are used to build the inventory and enable an authorized consumer to track resource changes over time. The O-Cloud provides a mechanism to discover the resource types with respective operations on resource types.

4.2.1.4.2.2 Attributes

Table 4.2.1.4.2.2-1 Attributes Properties for ResourceType

Attribute name	Data Type/Description	Properties

resourceTypeId	**Data Type:** string
	
	Description: Identifier of the resource type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Name of the resource type as identified by the cloud provider.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the resource type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

vendor	**Data Type:** string
	
	Description: Name of the provider of the resource type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

model	**Data Type:** string
	
	Description: Information about the model of the resource type
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

version	**Data Type:** string
	
	Description: Version or generation of the resource type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

alarmDictionaryId	**Data Type:** string
	
	Description: Identifier of the associated dictionary of alarms that can be generated by the resource type. If no alarms are reported against a resource of this resourceType then the
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`format: uuid`

Attribute name	Data Type/Description	Properties
	alarmDictionaryId is set to null.	

performanceDictionaryId	**Data Type:** string
	
	Description: Identifier of the associated dictionary of performance measurements that can be collected for the resource type. If the resource of this resourceType does not provide any measurements then the performanceDictionaryId is set to null.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`format: uuid`

resourceKind	**Data Type:** string
	
	Description: A value describing the "physicality" of the resource type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- UNDEFINED`
		`- PHYSICAL`
		`- LOGICAL`

resourceClass	**Data Type:** string
	
	Description: A value describing the functional role within the cloud for the resource type.
	
	The list of resource classes is not exhaustive, additional classes will be assessed on a case by case basis.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- UNDEFINED`
		`- COMPUTE`
		`- NETWORKING`
		`- STORAGE`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Resource Type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`
		`AttributeValuePair`

4.2.1.4.2.3 Attribute constraints

None.

4.2.1.4.2.4 State diagram

None.

4.2.1.4.3 ResourcePool

4.2.1.4.3.1 Definition

The ResourcePool represents a resource pool instance in an O-Cloud.

The O-Cloud is a distributed cloud with resources placed in different locations. Those resources are organized into resource pools. There can be more than one resource pool for a given O-Cloud at the same physical location. However, each resource pool requires a different identity since the SMO will direct a deployment based on homing policies to a specific location. This is different than typical cloud deployments as location is key to meeting latency requirements, sometime to elements outside of the O-Cloud.

NOTE 1: The identity of a resource pool cannot be tied to a specific resource within the pool as that resource may be replaced as part of the O-Cloud lifecycle.

NOTE 2: The resource pool is described in the Cloud Architecture and Deployment Secnarios Technical Report [i.3] but may not be needed for exposure through the Information Model and subsequent O2 IMS interface.

4.2.1.4.3.2 Attributes

Table 4.2.1.4.3.2-1 Attributes Properties for ResourcePool

Attribute name	Data Type/Description	Properties

resourcePoolId	**Data Type:** string
	
	Description: Identifier of the resource pool. Locally unique within the scope of an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Human readable name of the resource pool.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

oCloudSiteId	**Data Type:** string
	
	Description: Identifier of the O-Cloud site the resource pool is a part of.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

description	**Data Type:** string
	
	Description: Human readable description of the resource pool.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

resources	**Data Type:** array
	
	Description: List of resource IDs that are part of the resource pool.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items: string`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Resource Pool.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`

Attribute name	Data Type/Description	Properties
		AttributeValuePair

4.2.1.4.3.3 Attribute constraints

None.

4.2.1.4.3.4 State diagram

None.

4.2.1.4.4 Resource

4.2.1.4.4.1 Definition

The Resource represents an instance of a resource in the O-Cloud.

Resource pools are comprised of resources, wherein a specific instance of a resource can only be part of a single resource pool.

Resources are of a specific resource type. The resource type of a given resource instance does not change over the lifetime of such a resource instance.

The Resource is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The Resource inherits the attributes from the *InfrastructureInventoryObject_*.

4.2.1.4.4.2 Attributes

Table 4.2.1.4.4.2-1 Attributes Properties for Resource

Attribute name	Data Type/Description	Properties

resourceId	**Data Type:** string
	
	Description: Identifier of the resource. Locally unique within the scope of an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

description	**Data Type:** string
	
	Description: Human readable description of the resource.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

resourceTypeId	**Data Type:** string
	
	Description: Reference to the resource type of the resource.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

globalAssetId	**Data Type:** string
	
	Description: Identifier or serial number of the resource, if available. It is required only if the resource has been identified during its addition to the cloud as a reportable asset in the SMO
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

Attribute name	Data Type/Description	Properties
	inventory.	
resourcePoolId	**Data Type:** string **Description:** Reference to the resource pool that the resource is part of.	support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
elements	**Data Type:** array **Description:** The resource might be composed of smaller resources or other resource instances of a different type.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: Resource
tags	**Data Type:** array **Description:** Keywords describing or classifying the resource instance.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string
groups	**Data Type:** array **Description:** Keywords denoting groups a resource belongs to.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Resource.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.4.3 Attribute constraints

None.

4.2.1.4.4.4 State diagram

None.

4.2.1.4.5 DeploymentManager

4.2.1.4.5.1 Definition

The DeploymentManager represents an instance of a DMS in the O-Cloud. This class allows a client to discover the details about a DMS instance. This class provides a list of supported capabilities, and metrics for total capacity, current allocations, current reservations. For more information about DMS, refer to "O-RAN O2 General Aspects and Principles Specification, clause 1.3.1 "[i2].

The available capacity information supported by a DeploymentManager shall be calculated as Capacity - Allocated - Reserved.

4.2.1.4.5.2 Attributes

Table 4.2.1.4.5.2-1 Attributes Properties for DeploymentManager

Attribute name	Data Type/Description	Properties

deploymentManagerId	**Data Type:** string
	
	Description: Identifier of the deployment manager. Locally unique within the scope of an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Human readable name of the deployment manager as identified by the cloud provider.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the deployment manageras populated by the cloud provider.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

oCloudId	**Data Type:** string
	
	Description: Reference to the O-Cloud associated to the deployment manager.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

deploymentManagementServiceEndpoint	**Data Type:** string
	
	Description: Endpoint to access the management services of the deployment manager.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uri`

supportedLocations	**Data Type:** array
	
	Description: List of globalLocationIDs that were assigned to resource pools that resources allocated to the service instance belong to.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items: string`

capabilities	**Data Type:** array	`x-support-qualifier: M`

Attribute name	Data Type/Description	Properties
	Description: Information about the capabilities supported by this set of deployment management services based on the resources allocated to the service instance.	readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair
capacity	**Data Type:** array **Description:** Information about the available, allocated and reserved O-Cloud resources capacity allocated to the deployment manager.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Resource Type.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.5.3 Attribute constraints

None.

4.2.1.4.5.4 State diagram

None.

4.2.1.4.6 InventorySubscription

4.2.1.4.6.1 Definition

The InventorySubscription represents an instance of a subscription to infrastructure inventory notifications for when an *InfrastructureInventoryObject_* within the inventory changes.

Subscriptions are required for a notification to be generated. Information about the change is provided to the consumer when the consumers specified criteria is met. The absence of a filter means all events are sent.

4.2.1.4.6.2 Attributes

Table 4.2.1.4.6.2-1 Attributes Properties for InventorySubscription

Attribute name	Data Type/Description	Properties
subscriptionId	**Data Type:** string	x-support-qualifier: M

Attribute name	Data Type/Description	Properties

	Description: Identifier of the subscription. Locally unique within the scope of an O-Cloud instance.	`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

callback	**Data Type:** string
	
	Description: The URL of the callback to which notifications are delivered to.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uri`

consumerSubscriptionId	**Data Type:** string
	
	Description: The consumer may provide its identifier for tracking, routing, or identifying the subscription used to report the event.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

filter	**Data Type:** string
	
	Description: Criteria for events which do not need to be reported or will be filtered by the subscription notification service. Therefore, if a filter is not provided then all events are reported.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

4.2.1.4.6.3 Attribute constraints

None.

4.2.1.4.6.4 State diagram

None.

4.2.1.4.7 *InfrastructureInventoryObject_*

4.2.1.4.7.1 Definition

The *InfrastructureInventoryObject_* is an abstract class which shall be implemented by one of its specific sub-classes. The *InfrastructureInventoryObject_* does not have any common attributes.

The *InfrastructureInventoryObject_* is a Generalized Object for which specializations of this class, Specialized Object classes, can be included in a change notification.

4.2.1.4.7.2 Attributes

This class does not provide any attributes.

4.2.1.4.7.3 Attribute constraints

None.

4.2.1.4.7.4 State diagram

None.

4.2.1.4.8 InfrastructureInventoryEvent <<notification>>

4.2.1.4.8.1 Definition

The InfrastructureInventoryEvent is sent to consumers with an accepted subscription request and a filter criterion matching the content of the InfrastructureInventoryEvent attributes. This event can be triggered by change to an attribute in any subclass of the *InfrastructureInventoryObject_* in which the property for x-inventoryNotification: or x-stateChangeNotification is "True".

NOTE: It is up to the protocol and data model specification to determine the one or various protocol operations enabling the notification service to invoke the specified callback procedure.

Table 4.2.1.4.8.1-1 InfrastructureInventoryEvent Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-GEN13	*Optional clarification*
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-08	

The information representing InfrastructureInventoryEvent shall follow the provisions indicated in table 4.2.1.4.8.2-1.

4.2.1.4.8.2 Attributes

Table 4.2.1.4.8.2-1 Attributes Properties for InfrastructureInventoryEvent

Attribute name	Data Type/Description	Properties

consumerSubscriptionId	**Data Type:** string
	
	Description: The consumer may provide its identifier for tracking, routing, or identifying the subscription used to report the event.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

notificationEventType	**Data Type:** string
	
	Description: The type (CREATE, MODIFY, DELETE) of event being reported
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: enum`
		`enum:`
		`- CREATE`
		`- MODIFY`
		`- DELETE`

objectRef	**Data Type:** string
	
	Description: The resultant reference to the object on which the action occurred.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`format: url`

priorObjectState	**Data Type:** InfrastructureInventoryObject
	
	Description: If the Event type is 'MODIFY' or 'DELETE' this this field will be populated with a copy of the object prior to the
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

Attribute name	Data Type/Description	Properties
	event.	

postObjectState	**Data Type:** InfrastructureInventoryObject
	
	Description: If the Event type is 'CREATE' or 'MODIFY' this this field will be populated with a copy of the object after the event.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

4.2.1.4.8.3 Attribute constraints

None.

4.2.1.4.8.4 State diagram

None.

4.2.1.4.9 OCloudAvailableNotification <<notification>>

4.2.1.4.9.1 Definition

Although the Genesis lifecycle event precedes the existence of the O2ims in the O-Cloud there is a minimal set of data expected to be provided from the SMO to the process. This information is retained in the O-Cloud and is used to register the O-Cloud with the SMO once it becomes usable. Data expected to be included with the callback URI in the genesis data is the globalCloudId, each globalLocationId, and each globalAssetId for infrastructure at those locations. Once enough of the O-Cloud is discovered, provisioned and is available for additional provisioning or workload deployments, the O-Cloud shall send the O-CloudAvailableNotification to the consumer (SMO), thus registering its InfrastructureManagementServiceEndPoint with the SMO. After receiving the notification, the SMO will interogate the InfrastructureInventory APIs to account for corporate assets and discover the DMS service endpoints to allow workloads to be deployed.

NOTE: It is up to the protocol and data model specification to determine the one or various protocol operations enabling the notification service to invoke the specified callback procedure.

Table 4.2.1.4.9.1-1 OCloudAvailableNotification Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-GEN5	*Optional clarification*

The information representing OCloudAvailableNotification shall follow the provisions indicated in table 4.2.1.4.9.2-1.

4.2.1.4.9.2 Attributes

Table 4.2.1.4.9.2-1 Attributes Properties for OCloudAvailableNotification

Attribute name	Data Type/Description	Properties

globalCloudId	**Data Type:** string
	
	Description: Identifier of the O-Cloud instance assigned by the SMO. This identifier is globally unique across O-Cloud instances known to the SMO. This value was provided by the SMO at cloud genesis and is stored in the O-Cloud IMS Inventory.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

Attribute name	Data Type/Description	Properties

oCloudId	**Data Type:** string
	
	Description: Identifier of the O-Cloud instance. Unique within an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

IMS_EP	**Data Type:** string
	
	Description: URL to the resource within the O-Cloud for other attributes.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uri`

4.2.1.4.9.3 Attribute constraints

None.

4.2.1.4.9.4 State diagram

None.

4.2.1.4.10 AlarmEventRecord <<dataType>>

4.2.1.4.10.1 Definition

The AlarmEventRecord represents a stored instance of an alarm detected within the O-Cloud Resources and stored by the IMS.

The AlarmEventRecord can represent either a current active alarm or one that occurred in the past. Retention of an AlarmEventRecord is dependent on IMS configuration which would be set through an IMS Provisioning service.

Table 4.2.1.4.10.1-1 AlarmEventRecord Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-17	*Optional clarification*
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-73	

4.2.1.4.10.2 Attributes

Table 4.2.1.4.10.2-1 Attributes Properties for AlarmEventRecord

Attribute name	Data Type/Description	Properties

alarmEventRecordId	**Data Type:** string
	
	Description: Identifier of an entry in the AlarmEventRecord. Locally unique within the scope of an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

objectTypeId	**Data Type:** string
	
	Description: A reference to the type of resource which caused the alarm. This could be from any namespace (e.g., a resourceTypeId in the inventory namespace, or a nodeClusterTypeId in the cluster
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

Attribute name	Data Type/Description	Properties
	namespace). The objectClass attribute identifies which class of object is being referenced.	

objectId	**Data Type:** string
	
	Description: A reference to the resource instance which caused the alarm. This could be from any namespace (e.g., a resourceId in the inventory namespace, or a nodeClusterId in the cluster namespace). The objectClass attributes identifies which class of object is being referenced.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

objectClass	**Data Type**: string
	
	Description: The name of the class of the object being referenced by the objectId attribute. This includes the namespace and class name of the object being referenced to disambiguate it from other objects in the same namespace as well as objects in other namespaces. From this a client can infer the type of objectType referenced by the objectTypeId attribute.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`example:`
		`“oran.o2ims.cluster.ClusterResource”`

alarmDefinitionId	**Data Type:** string
	
	Description: A reference to the Alarm Definition record in the Alarm Dictionary associated with the referenced Object Type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

probableCauseId	**Data Type:** string
	
	Description: A reference to the ProbableCause of the Alarm.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

alarmRaisedTime	**Data Type:** string
	
	Description: This field is populated with a Date/Time stamp value when the AlarmEventRecord is created.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: date-time`

alarmChangedTime	**Data Type:** string
	
	Description: This field is populated with a Date/Time stamp value when any value of the AlarmEventRecord is modified.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: date-time`

alarmClearedTime	**Data Type:** string
	
	Description: This field is
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`

Attribute name	Data Type/Description	Properties
	populated with a Date/Time stamp value when the alarm condition is cleared.	x-inventoryNotification: False x-stateChangeNotification: True nullable: True format: date-time
alarmAcknowledgeTime	**Data Type:** string **Description:** This field is populated with a Date/Time stamp value when the alarm condition is acknowledged.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: True format: date-time
alarmAcklnowledged	**Data Type:** boolean **Description:** This is a Boolean value defaulted to False. When a system acknowledges an alarm, it is then set to True.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True default: False
perceivedSeverity	**Data Type:** string **Description:** This is an enumerated set of values which identify the perceived severity of the alarm.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False enum: - CRITICAL - MAJOR - MINOR - WARNING - INDETERMINATE - CLEARED
extensions	**Data Type:** array **Description:** Unspecified (not standardized) properties (keys) which are tailored by the vendor or operator to extend the information provided about the O-Cloud Alarm.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.10.3 Attribute constraints

None.

4.2.1.4.10.4 State diagram

None.

4.2.1.4.11 AlarmEvent <<notification>>

4.2.1.4.11.1 Definition

The AlarmEvent represents a notification of an Alarm sent by the IMS to a registered Subcriber for a fault detected within an O-Cloud Resources.

The AlarmEvent can represent either a new alarm, a change of occurrence, or state change of an existing alarm.

Table 4.2.1.4.11.1-1 AlarmEvent Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-16	*Optional clarification*

4.2.1.4.11.2 Attributes

Table 4.2.1.4.11.2-1 Attributes Properties for AlarmEvent

Attribute name	Data Type/Description	Properties

globalCloudId	**Data Type:** string
	
	Description: Identifier of the O-Cloud instance assigned by the SMO. This identifier is globally unique across O-Cloud instances known to the SMO. This value was provided by the SMO at cloud genesis and is stored in the O-Cloud IMS Inventory.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerSubscriptionID	**Data Type:** string
	
	Description: Identifier provided by the subscriber for event to subscription traceability within the consumer realm.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

alarmNotificationType	**Data Type:** string
	
	Description: This is an enumerated set of values which identify the type of notification being sent to the subscriber.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- NEW`
		`- CHANGE`
		`- CLEAR`
		`- ACKNOWLEDGE`

objectRef	**Data Type:** string
	
	Description: This is a URL entry to the AlarmEventRecord for this notification within the AlarmList
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: url`

alarmEventRecord	**Data Type:** AlarmEventRecord
	
	Description: This is a copy of the AlarmEventRecord associated with this event preventing the need for the alarm consumer to do a read after the notification is received.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

4.2.1.4.11.3 Attribute constraints

None.

4.2.1.4.11.4 State diagram

None.

4.2.1.4.12 AlarmSubscription

4.2.1.4.12.1 Definition

The AlarmSubscription represents a consumer wishing to be notified when specified alarm conditions occur.

The AlarmSubscription can be filtered by criteria based on the type of notification of fields of the AlarmEventRecord.

Table 4.2.1.4.12.1-1 Subscription Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-14	
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-15	

4.2.1.4.12.2 Attributes

Table 4.2.1.4.12.2-1 Attributes Properties for AlarmSubscription

Attribute name	Data Type/Description	Properties

alarmSubscriptionId	**Data Type:** string
	
	Description: Identifier of the subscription. Locally unique within the scope of an O-Cloud instance.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerSubscriptionID	**Data Type:** string
	
	Description: The consumer may provide its identifier for tracking, routing, or identifying the subscription used to report the event if a value was provided in the SubscriptionRequest.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

filter	**Data Type:** string
	
	Description: Criteria for events which do not need to be reported or will be filtered by the subscription notification service. Therefore, if a filter is not provided then all events are reported.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`  - NEW`
		`  - CHANGE`
		`  - CLEAR`
		`  - ACKNOWLEDGE`

callback	**Data Type:** string
	
	Description: The URI of the callback to which notifications are delivered to.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: url`

4.2.1.4.12.3 Attribute constraints

None.

4.2.1.4.12.4 State diagram

None.

4.2.1.4.13 AlarmList

4.2.1.4.13.1 Definition

The AlarmList represents the configuration parameters of the AlarmList and the list of AlarmEventRecords currently maintained in the O-Cloud.

Table 4.2.1.4.13.1-1 AlarmList Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	[REQ-ORC-O2-74]	
O-RAN.WG6.ORCH-USE-CASES [3]	[REQ-ORC-O2-98]	

4.2.1.4.13.2 Attributes

Table 4.2.1.4.13.2-1 Attributes Properties for AlarmList

Attribute name	Data Type/Description	Properties
retentionPeriod	**Data Type:** integer **Description:** Specifies the number of days to retain cleared/acknowledge alarms in the AlarmList. Entries in the alarmEventRecords will be at least as long as the retentionPeriod but can be longer depending on the implementation of the purge feature.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False x-units: Days minimum: 0
alarmEventRecords	**Data Type:** array **Description:** The alarmEventRecords provides the storage array for AlarmEventRecords that are active or have been cleared but not purged according to the implementation of the purge feature.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: True nullable: False minItems: 0 uniqueItems: True items: AlarmEventRecord
extensions	**Data Type:** array **Description:** These are unspecified properties (keys and values) which can be tailored to control the behaviour of the Alarm List.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.13.3 Attribute constraints

None.

4.2.1.4.13.4 State diagram

None.

4.2.1.4.14 PerformanceMeasurementStore

4.2.1.4.14.1 Definition

The PerformanceMeasurementStore represents the O-Cloud internal store where PerformanceMeasurementRecord(s) are retained according to the retention policy of the O-Cloud based on the configured retention period.

Table 4.2.1.4.14.1-1 PerformanceMeasurementStore Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	[REQ-ORC-O2-100]	

The information representing the attributes of the PerformanceMeasurementStore follows the provisions indicated in table 4.2.1.4.14.2-1.

4.2.1.4.14.2 Attributes

Table 4.2.1.4.14.2-1 Attributes Properties for PerformanceMeasurementStore

Attribute name	Data Type/Description	Properties

retentionPeriod	**Data Type:** integer
	
	Description: The number in days (24 hours) that measurements will be retained. The exact timing of when measurements are deleted are implementation dependent.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uInteger`
		`x-units: days`
		`minimum: 0`

measurements	**Data Type:** array
	
	Description: List of PerformanceMeasurementRecord(s) collected and stored in the performance measurement store.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`
		`PerformanceMeasurementRecord`

extensions	**Data Type:** array
	
	Description: These are unspecified properties (keys and values) which can be tailored to control the behaviour of the Performance Service.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`
		`AttributeValuePair`

4.2.1.4.14.3 Attribute constraints

None.

4.2.1.4.14.4 State diagram

None.

4.2.1.4.15 PerformanceMeasurementRecord <<dataType>>

4.2.1.4.15.1 Definition

The PerformanceMeasurementRecord represents a collected measurement in the PerformanceMeasurementStore.

The information representing the attributes of the PerformanceMeasurementRecord follows the provisions indicated in table 4.2.1.4.15.2-1.

4.2.1.4.15.2 Attributes

Table 4.2.1.4.15.2-1 Attributes Properties for PerformanceMeasurementRecord

Attribute name	Data Type/Description	Properties

performanceMeasurementJobId	**Data Type:** string
	
	Description: The job identifier assigned by the O-Cloud when a performance measurement job is created.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

resourceId	**Data Type:** string
	
	Description: The resource identifier for the resource for which the collected measurement is against.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

performanceMeasurementDefinitionId	**Data Type:** string
	
	Description: The Performance Dictionary definition identifier for the collected measurement.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

timeStamp	**Data Type:** string
	
	Description: The date and time in which the measurement was collected.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: date-time`

measurementValue	**Data Type:** Any type
	
	Description: The value of the measurement collected. The ValueType represents a variety of types depending on the measurement type.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

isSuspect	**Data Type:** boolean
	
	Description: This flag is set to True if anomalous behaviour was detected on the resource during the last collection interval.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

4.2.1.4.15.3 Attribute constraints

None.

4.2.1.4.15.4 State diagram

None.

4.2.1.4.16 PerformanceMeasurementJob

4.2.1.4.16.1 Definition

The PerformanceMeasurementJob represents a process established to collect measurements and place them in the PerformanceMeasurementStore.

The information representing the attributes of the PerformanceMeasurementJob follows the provisions indicated in table 4.2.1.4.16.2-1.

For attributes with x-inventoryNotification: set to True or x-stateChangeNotification set to True, the notification to be sent will be the PerformanceMeasurementJobChangeNotification.

Table 4.2.1.4.16.1-1 PerformanceMeasurementJob Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-31	*Optional clarification*
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-32	

4.2.1.4.16.2 Attributes

Table 4.2.1.4.16.2-1 Attributes Properties for PerformanceMeasurementJob

Attribute name	Data Type/Description	Properties

performanceMeasurementJobId	**Data Type:** string
	
	Description: Identifier of the performance job instance within the IMS. This value is assigned by the IMS when the job is created
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerPerformanceJobId	**Data Type:** string
	
	Description: Identifier optionally provided by the consumer for its purpose of managing performance jobs. This value could be the same across multiple instances of IMS performance job instances.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`

state	**Data Type:** string
	
	Description: The current state of the PerformanceMeasurementJob.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: True`
		`nullable: False`
		`enum:`
		`- ACTIVE`
		`- SUSPENDED`
		`- DEPRECATED`

collectionInterval	**Data Type:** integer
	
	Description: The interval at which performance measurements will be collected
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-units: Seconds`

Attribute name	Data Type/Description	Properties
	and stored.	minimum: 0 default: 300
resourceScopeCriteria	**Data Type:** array **Description:** Key value pairs of resource attributes which are used to select resources from which measures are to be collected. This criterion determines, the value set for the Qualified Resource Types.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair example: [{"resourceTypeId","["7c491f8f-7207-4c00-9b67-3d2ee8b008f0", "31040dec-8106-44db-83bc-62e1d618ea17"]"}]
measurementSelectionCriteria	**Data Type**: array **Description**: Key value pairs that identify the distinct set of measurements within the scope of a PerformanceMeasurementJob.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: AttributeValuePair example: [{"measurementGroup","MemoryUsage"}, {"measurementName","cpuAverageUtilization"}]
status	**Data Type:** string **Description:** The current status within the state.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False enum: - RUNNING - FAILED - DEGRADED - IDLE - PENDING_DELETE
preinstalledJob	**Data Type:** boolean **Description:** This is an indicator if the performance job was created for the IMS (True) or as the result of an IMS consumer request (False)	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
qualifiedResourceTypes	**Data Type:** array	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 0 uniqueItems: True items: uuid example:'["7c491f8f-7207-4c00-9b67-3d2ee8b008f0", "31040dec-8106-44db-83bc-62e1d618ea17"]'

Attribute name	Data Type/Description	Properties
	Description: This list is computed from evaluation of the resourceScopeCriteria. The resulting qualifiedResourceTypes are the distinct set of ResourceTypes among those measuredResources. This list of resource type UUIDs is used to qualify collectedMeasurements defined in the PM Dictionary of the qualified resource types.	

measuredResources	**Data Type:** array
	
	Description: This is the historical list of resources measured by this job. Resources added and deleted are kept track of for the life of the job.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: MeasuredResource`

collectedMeasurements	**Data Type:** array
	
	Description: The historical list of measurements that are or have been collected by this job based on its resourceScopeCriteria and measurementSelectionCriteria over the life of the job.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: CollectedMeasurement`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (key/value) which extend the information about the Performance Measurement Job.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: AttributeValuePair`

4.2.1.4.16.3 Attribute constraints

Table 4.2.1.4.16.3-1 Attributes Constraints for PerformanceMeasurementJob

Name	Definition
Status is RUNNING, FAILED, or DEGRADED	Condition: Job is in the ACTIVE state.
Status is IDLE	Condition: Job is in the SUSPENDED state.
Status is PENDING_DELETE	Condition: Job is in the DEPRECATED state.
WriteableExtensions	Condition: preInstalledJob is False then extensions can be created and updated.
ReadOnlyExtensions	Condition: preInstalledJob is True then extensions, if present are read only.

4.2.1.4.16.4 State diagram

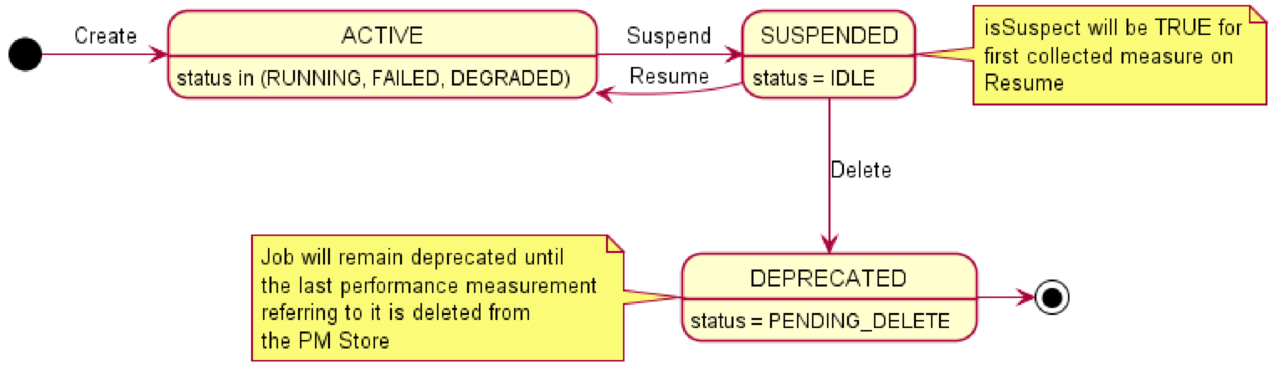

Figure 4.2.1.4.16.4-1 State & Status Diagram for PerformanceMeasurementJob

4.2.1.4.17 MeasuredResource <<datatype>>

4.2.1.4.17.1 Definition

The MeasuredResource represents a Resource that is or has been within the scope of a PerformanceMeasurementJob.

The information representing the attributes of the MeasuredResource follows the provisions indicated in table 4.2.1.4.17.2-1.

4.2.1.4.17.2 Attributes

Table 4.2.1.4.17.2-1 Attributes Properties for MeasuredResource

Attribute name	Data Type/Description	Properties
resourceTypeId	**Data Type:** string **Description:** The resource identifier assigned by the O-Cloud when a performance measurement job is created.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification:False nullable: False format: uuid
resourceId	**Data Type:** string **Description:** The resource identifier for the resource for which the collected measurement is against.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
timeAdded	**Data Type:** array **Description:** List of timestamps when the resource was added to the job. All resources shall have at least one timeAdded value. It could have multiple if changes caused a deletion and subsequent reinstatement to occur.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 uniqueItems: True items: date-time
timeDeleted	**Data Type:** array **Description:** List of timestamps when the resource was deleted from a job. Multiple values may be present if deletion occurred at different times. The effective	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minItems: 0 uniqueItems: True

Attribute name	Data Type/Description	Properties
	period of the measurement would be the times between a timeAdded and timeDeleted values.	items: date-time
isCurrentlyMeasured	**Data Type:** boolean **Description:** Indicates if the resource is currently being measured by the contained performance job.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False

4.2.1.4.17.3 Attribute constraints

None.

4.2.1.4.17.4 State diagram

None.

4.2.1.4.18 CollectedMeasurement <<datatype>>

4.2.1.4.18.1 Definition

The CollectedMeasurement represents a measurement associated with a specific ResourceType that is or has been within the scope of a PerformanceMeasurementJob.

The information representing the attributes of the CollectedMeasurement follows the provisions indicated in table 4.2.1.4.18.2-1.

4.2.1.4.18.2 Attributes

Table 4.2.1.4.18.2-1 Attributes Properties for CollectedMeasurement

Attribute name	Data Type/Description	Properties
resourceTypeId	**Data Type:** string **Description:** The resource identifier assigned by the O-Cloud when a performance measurement job is created.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
performanceMeasurementDefinitionId	**Data Type:** string **Description:** Identifier of the measurement definition with the PM Dictionary for the specific device type identified by the resourceTypeId.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
timeAdded	**Data Type:** array **Description:** List of timestamps when the resource was added to the job. All measurements shall have at least one timeAdded value. It could have multiple if changes caused a deletion and subsequent reinstatement to	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 uniqueItems: True items: date-time

Attribute name	Data Type/Description	Properties
	occur.	

timeDeleted	**Data Type:** array
	
	Description: List of timestamps when the measurement was deleted from a job. Multiple values may be present if deletion occurred at different times. The effective period of the measurement would be the times between a timeAdded and timeDeleted values.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`uniqueItems: True`
		`items: date-time`

isCurrentlyMeasured	**Data Type:** boolean
	
	Description: Indicates if the resource is currently being measured by the contained performance job.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

4.2.1.4.18.3 Attribute constraints

None.

4.2.1.4.18.4 State diagram

None.

4.2.1.4.19 PerformanceMeasurementJobChangeSubscription

4.2.1.4.19.1 Definition

The PerformanceMeasurementJobChangeSubscription represents a subscription for events to be notified such as when the scope of a PerformanceMeasurementJob changes either due to an update to its criteria or a change in inventory, or updates to other attributes (e.g. collection interval, state, status).

The information representing the attributes of the PerformanceJobChangeSubscription follows the provisions indicated in table 4.2.1.4.19.2-1.

4.2.1.4.19.2 Attributes

Table 4.2.1.4.19.2-1 Attributes Properties for PerformanceMeasurementJobChangeSubscription

Attribute name	Data Type/Description	Properties

performanceMeasurementJobChangeSubscriptionId	**Data Type:** string
	
	Description: Identifier of the PerformanceMeasurementJobChangeSubscription which will generate this notification.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerPerformanceMeasurementJobChangeSubscriptionId	**Data Type:** string
	
	Description: The consumer identification value if provided on the PerformanceMeasurementJobChangeSubscription which will
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`

Attribute name	Data Type/Description	Properties
	generate this notification.	
performanceChangesToNotify	**Data Type:** string **Description:** The enumerated set of values representing the type(s) of changes to resources and/or measurements desired to be notified about (ADD, DELETE, BOTH).	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False enum: - ADD - DELETE - BOTH
detailsToInclude	**Data Type:** string **Description:** The enumerated set of values representing how much detail about the change to to resources and/or measurements include in the notification (NONE, DELTA, COMPLETE).	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False enum: - NONE - DELTA - COMPLETE
subscriptionJobFilter	**Data Type:** array **Description:** A key value set of criteria to select which jobs shall be reported on.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False x-isOrdered: True minItems: 0 uniqueItems: True items: AttributeValuePair
callback	**Data Type:** string **Description:** URL to which the PerformanceMeasurementJobChangeNotification will be sent.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uri

4.2.1.4.19.3 Attribute constraints

None.

4.2.1.4.19.4 State diagram

None.

4.2.1.4.20 PerformanceMeasurementJobChangeNotification <<notification>>

4.2.1.4.20.1 Definition

The PerformanceMeasurementJobChangeNotification notification represents the information about a change that occurred to a PerformanceMeasurementJob due to an update to its criteria or a change in inventory or updates to other attributes (e.g. collection interval, state, status) for which there is a PerformanceMeasurementJobChangeSubscription established.

The information representing the attributes of the PerformanceMeasurementJobChangeNotification follows the provisions indicated in table 4.2.1.4.20.2-1.

4.2.1.4.20.2 Attributes

Table 4.2.1.4.20.2-1 Attributes Properties for PerformanceMeasurementJobChangeNotification

Attribute name	Data Type/Description	Properties

performanceMeasurementJobChangeSubscriptionId	**Data Type:** string
	
	Description: Identifier of the PerformanceMeasurementJobChangeSubscription.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerPerformanceMeasurementJobChangeSubscriptionId	**Data Type:** string
	
	Description: The consumer identification value if provided on the PerformanceMeasurementJobChangeSubscription.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

performanceMeasurementJobId	**Data Type:** string
	
	Description: The UUID of the PerformanceMeasurementJob that incurred a change due to a change in scope criteria, a change to O-Cloud Infrastructure Inventory, a change to collection interval, a change to state and/or status, or a change to the extensions.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

objectRef	**Data Type:** string
	
	Description: The URL of the service object which was modified.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uri`

typeOfChange	**Data Type:** string
	
	Description: The enumerated value representing the type of changes to resources and/or measurements included in the notification.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- ADD`
		`- DELETE`
		`- BOTH`

resourceChangeDetails	**Data Type:** array
	
	Description: If the subscription criteria indicates DELTA or COMPLETE extended information then all relevant added and/or deleted resources to the job are listed.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`uniqueItems: True`
		`x-isOrdered: True`
		`items: MeasuredResource`

measurementChangeDetails	**Data Type:** array	`x-support-qualifier: M`

Attribute name	Data Type/Description	Properties
	Description: If the subscription criteria indicates DELTA or COMPLETE extended information then all relevant added and/or deleted measurements of the performance measurement job are listed.	readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minItems: 0 uniqueItems: True x-isOrdered: True items: CollectedMeasurement
consumerPerformanceJobId	**Data Type:** string **Description:** Identifier provided by the consumer for its purpose of managing performance jobs. This value could be the same across multiple instances of IMS performance job instances.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True
collectionInterval	**Data Type:** integer **Description:** The interval at which performance measurement will be collected and stored.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-units: Seconds minimum: 0 default: 300
state	**Data Type:** string **Description:** The current state of the PerformanceMeasurementJob.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True enum: - ACTIVE - SUSPENDED - DEPRECATED
status	**Data Type:** string **Description:** The current status within the state.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True enum: - RUNNING - FAILED - DEGRADED - IDLE - PENDING_DELETE
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (key/value) which extend the information about the Performance Measurement Job	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.20.3 Attribute constraints

None.

4.2.1.4.20.4 State diagram

None.

4.2.1.4.21 PerformanceSubscription

4.2.1.4.21.1 Definition

The PerformanceSubscription represents the information required to establish the criteria by which a PerformanceMeasurementReport notification is to be generated and where to send that notification.

Table 4.2.1.4.21.1-1 PerformanceSubscription Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-37	*Optional clarification*
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-38	

The information representing the attributes of the PerformanceSubscription follows the provisions indicated in table 4.2.1.4.21.2-1.

4.2.1.4.21.2 Attributes

Table 4.2.1.4.21.2-1 Attributes Properties for PerformanceSubscription

Attribute name	Data Type/Description	Properties

performanceSubscriptionId	**Data Type:** string
	
	Description: Identifier of the performance subscription instance within the IMS. This value is assigned by the IMS when the subscription is created.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerPerformanceSubscriptionId	**Data Type:** string
	
	Description: Identifier optionally provided by the consumer for its purpose of managing performance data. This value could be the same across multiple instances of IMS subscription instances.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

globalSubscriptionCriteria	**Data Type:** array
	
	Description: This is subscription criteria which will apply to each contained subscription.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`
		
		`PerformanceSubscriptionCriteria`

reportFormat	**Data Type:** string	`x-support-qualifier: M`

Attribute name	Data Type/Description	Properties

	Description: Enumerated set of reporting mechanism for delivery of the data identified by the PerformanceSubscription which are identified as:
	NOTIFICATION: Call the callback with the data at the specified intervals providing the data, e.g., in JSON format
	FILE: generate a file, e.g., formatted in XML to be collected and call the callback with the filename whenever one is generated.
	STREAM:Session will remain open and continue to send the data in a streaming format, e.g., Google Protocol Buffers (GPB), at the specified intervals until the session or subscription is terminated.
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- NOTIFICATION`
		`- FILE`
		`- STREAM`

remoteFileLocation	**Data Type:** string
	Description: When the reportFormat is FILE this field is populated with the URL to the remote directory where the service is expected to push the requested performance files to a remote files system. If the reportFormat is not FILE, the field is empty or not present.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`Format: uri`

callback	**Data Type:** string
	Description: The callback to send responses for modes NOTIFICATION, or the file notification for FILE. The field is not required if mode is STREAM.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`Format: uri`

measurementReportingFrequencies	**Data Type:** array
	Description: The frequency that specified measurements are to be reported.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items:`
		`PerformanceReportingFrequency`

4.2.1.4.21.3 Attribute constraints

None.

4.2.1.4.21.4 State diagram

None.

4.2.1.4.22 PerformanceReportingFrequency <<datatype>>

4.2.1.4.22.1 Definition

The PerformanceReportingFrequency represents the frequency that specified measurements are to be reported.

The information representing the attributes of the PerformanceReportingFrequency follows the provisions indicated in table 4.2.1.4.22.2-1.

4.2.1.4.22.2 Attributes

Table 4.2.1.4.22.2-1 Attributes Properties for PerformanceReportingFrequency

Attribute name	Data Type/Description	Properties

subscriptionCriteria	**Data Type:** PerformanceSubscriptionCriteria
	
	Description: This is subscription criteria which will apply to this subscription.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

subscriptionMode	**Data Type**: string
	
	Description: An enumerated set of values which will identify the mode in which data is reported. The valid values are as follows:
	TARGET_DEFINED: Producer of the PM data decides if ON_CHANGE or SAMPLE is more optimal.
	ON_CHANGE: Value is reported as soon as the PM Job collects it.
	SAMPLE: Value is reported at the reportInterval period.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`enum:`
		`- TARGET_DEFINED`
		`- ON_CHANGE`
		`- SAMPLE`

reportInterval	**Data Type:** integer
	
	Description: This is the reporting interval for the data, it is an unsigned integer representing the number of milliseconds between reports. If the reportInterval is lower than the service can provide the subscription shall return an error indicating such. If the value is set to zero (0) then the service will use its lowest supported value for the reportInterval.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minimum: 0`
		`default: 0`
		`x-units: milliseconds`

Attribute name	Data Type/Description	Properties
supressRedundant	**Data Type:** boolean **Description:** This flag identifying if SAMPLE values that have not changed shall (False) or shall Not (True) be transmitted.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True default: False
heartbeatInterval	**Data Type:** integer **Description:** When the suppressRedundant is set to (True), if heartbeatInterval is a non-zero value then a duplicate value will be sent at least once per heatbeatInterval. The value shall be zero (0) or a value greater than or equal to the reportInterval.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minimum: 0 default: 0 x-units: milliseconds

4.2.1.4.22.3 Attribute constraints

None.

4.2.1.4.22.4 State diagram

None.

4.2.1.4.23 PerformanceSubscriptionCriteria <<datatype>>

4.2.1.4.23.1 Definition

The PerformanceSubscriptionCriteria represents the information required to establish which measurements for which resource shall be included in a PerformanceMeasurementReport.

The information representing the attributes of the PerformanceSubscriptionCriteria follows the provisions indicated in table 4.2.1.4.23.2-1.

4.2.1.4.23.2 Attributes

Table 4.2.1.4.23.2-1 Attributes Properties for PerformanceSubscriptionCriteria

Attribute name	Data Type/Description	Properties
jobCriteria	**Data Type:** array **Description:** Key/Value list of criteria compared to Job data.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minItems: 0 uniqueItems: True items: AttributeValuePair
resourceTypeCriteria	**Data Type:** array **Description:** Key/Value list of criteria compared to Resource Type data.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minItems: 0 uniqueItems: True items:

Attribute name	Data Type/Description	Properties
		AttributeValuePair
resourceCriteria	**Data Type:** array **Description:** Key/Value list of criteria compared to Resource data.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True minItems: 0 uniqueItems: True items: AttributeValuePair
measurementCriteria	**Data Type:** array **Description:** Key/Value list of criteria compared to Measurement Definition data.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification:False nullable: True minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.23.3 Attribute constraints

None.

4.2.1.4.23.4 State diagram

None.

4.2.1.4.24 PerformanceMeasurementReport <<notification>>

4.2.1.4.24.1 Definition

The PerformanceMeasurementReport notification represents the information required to define the scope of the JobReports being sent in the notification.

Table 4.2.1.4.24.1-1 PerformanceMeasurementReport Requirements

Referenced Specification	Requirement	Comment
O-RAN.WG6.ORCH-USE-CASES [3]	REQ-ORC-O2-39	*Optional clarification*

The information representing the attributes of the PerformanceMeasurementReport follows the provisions indicated in table 4.2.1.4.24.2-1.

4.2.1.4.24.2 Attributes

Table 4.2.1.4.24.2-1 Attributes Properties for PerformanceMeasurementReport

Attribute name	Data Type/Description	Properties
globalCloudId	**Data Type:** string **Description:** Identifier of the O-Cloud instance assigned by the SMO. This identifier is globally	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False

Attribute name	Data Type/Description	Properties

	unique across O-Cloud instances known to the SMO. This value was provided by the SMO at cloud genesis and is stored in the O-Cloud IMS Inventory.	`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

notificationTime	**Data Type:** string
	
	Description: The date and time that the notification was sent.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: date-time`

performanceSubscriptionId	**Data Type:** string
	
	Description: Identifier of the performance subscription instance within the IMS. This value is assigned by the IMS when the job is created.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerPerformanceSubscriptionId	**Data Type:** string
	
	Description: Identifier optionally provided by the consumer for its purpose of managing performance data. This value could be the same across multiple instances of IMS subscription instances.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

jobReports	**Data Type:** array
	
	Description: The list of performance reports by jobId. A single subscription can subscribe to measurements collected by multiple jobs.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items: JobReport`

4.2.1.4.24.3 Attribute constraints

None.

4.2.1.4.24.4 State diagram

None.

4.2.1.4.25 JobReport <<datatype>>

4.2.1.4.25.1 Definition

The JobReport provides the list of performance reports generated from corresponding PerformanceMeasurementJobs. A single subscription can be used to subscribe to measurements collected by multiple PerformanceManagementJobs. The data from the PerformanceMeasurementJobs are aggregated into a single report per PerformanceSubscription.

The information representing the attributes of the JobReport follows the provisions indicated in table 4.2.1.4.25.2-1.

4.2.1.4.25.2 Attributes

Table 4.2.1.4.25.2-1 Attributes Properties for JobReport

Attribute name	Data Type/Description	Properties

performanceJobId	**Data Type:** string
	
	Description: The UUID of the PerformanceMeasurementJob from which collected measurements are reported.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

consumerJobId	**Data Type:** string
	
	Description: Identifier provided by the consumer for its purpose of managing performance jobs. This value could be the same across multiple instances of IMS performance job instances.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

measuredResources	**Data Type:** array
	
	Description: The list of collected measurements for each Resource within the scope of the subscription to be included in this PerformanceMeasurementRepor t.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`minItems: 1`
		`items: ResourceReport`

4.2.1.4.25.3 Attribute constraints

None.

4.2.1.4.25.4 State diagram

None.

4.2.1.4.26 ResourceReport <<datatype>>

4.2.1.4.26.1 Definition

The ResourceReport dataType represents the information required to identify the resource for which a set of measurements are being reported in the notification.

The information representing the attributes of the ResourceReport follows the provisions indicated in table 4.2.1.4.26.2-1.

4.2.1.4.26.2 Attributes

Table 4.2.1.4.26.2-1 Attributes Properties for ResourceReport

Attribute name	Data Type/Description	Properties

resourceId	**Data Type:** string
	
	Description: The UUID of the Resource as specified in Infrastructure Inventory.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

measurementValues	**Data Type:** array	`x-support-qualifier: M` `readOnly: True`

Attribute name	Data Type/Description	Properties
	Description: The list of collected measurements for the Resource within the subscription included in this PerformanceMeasurementReport.	x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 uniqueItems: True items: MeasurementValue

4.2.1.4.26.3 Attribute constraints

None.

4.2.1.4.26.4 State diagram

None.

4.2.1.4.27 MeasurementValue <<datatype>>

4.2.1.4.27.1 Definition

The MeasurementValue dataType represents the information about a collected measurement and its value to be included in a notification.

The information representing the attributes of the MeasurementValue follows the provisions indicated in table 4.2.1.4.27.2-1.

4.2.1.4.27.2 Attributes

Table 4.2.1.4.27.2-1 Attributes Properties for MeasurementValue

Attribute name	Data Type/Description	Properties
performanceMeasurementId	**Data Type:** string **Description:** Based on encoding this value can be either a string consisting of the performanceMeasureName of the MeasurementDefinition in the Performance Dictionary for the Resource Type or an unsigned integer consisting of the performanceMeasurementId of the MeasurementDefinition.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
measurementCollectionTime	**Data Type:** string **Description:** If present, this is the value that is added to the PerformanceMeasurementStore. When omitted then the notificationTime is assumed to the time the value was added to the PerformanceMeasurementStore as would be the case if the subscriptionMode equated to ON_CHANGE.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True format: date-time
measurementValue	**Data Type:** Any type **Description:** This is the actual value of the measurement which varies in type from measurement	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False

Attribute name	Data Type/Description	Properties
	to measurement.	nullable: False
isSuspect	**Data Type:** boolean **Description:** This flag is set to True if anomalous behavior was detected on the resource between this measurement value and a prior measurement value.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True default: False

4.2.1.4.27.3 Attribute constraints

None.

4.2.1.4.27.4 State diagram

None.

4.2.1.4.28 NotifyFileReady <<notification>>

4.2.1.4.28.1 Definition

The NotifyFileReady notification represents the information about one (1) or more files to be made available to a subscriber when the reportFormat of a PerformanceSubscription is FILE or a LogQuery when a callback is supplied. This notification only makes the consumer aware of the file and does not predicate delivery of the file (push or pull).

The information representing the attributes of the NotifyFileReady follows the provisions indicated in table 4.2.1.4.28.2-1.

4.2.1.4.28.2 Attributes

Table 4.2.1.4.28.2-1 Attributes Properties for NotifyFileReady

Attribute name	Data Type/Description	Properties
globalCloudId	**Data Type:** string **Description:** Identifier of the O-Cloud instance assigned by the SMO. This identifier is globally unique across O-Cloud instances known to the SMO. This value was provided by the SMO at cloud genesis and is stored in the O-Cloud IMS Inventory.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
subscriptionId	**Data Type:** string **Description:** When the notification is created due to a PerformanceSubscription this attribute will contain the identifier of the performanceSubscriptionId for this notification.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: True format: uuid
consumerSubscriptionId	**Data Type:** string **Description:** When the notification is created due to a PerformanceSubscription this attribute will contain the identifier provided by the consumer for its purpose of managing performance data and may be	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True

Attribute name	Data Type/Description	Properties
	included in the PerformanceSubscription. If the notification is due to a logQuery request this attribute shall provide the consumerLogRequestId when provided on the request to allow the consumer to correlate it to the original request.	
eventTime	**Data Type:** string **Description:** The date and time that the notification was issued.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: date-time
fileInfoList	**Data Type:** array **Description:** The list of information about one or more files included in this notification.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 uniqueItems: True items: FileInfo

4.2.1.4.28.3 Attribute constraints

None.

4.2.1.4.28.4 State diagram

None.

4.2.1.4.29 FileInfo <<datatype>>

4.2.1.4.29.1 Definition

The FileInfo dataType represents information about one or more generated files collected by a producer and which are now available to a consumer.

The information representing the attributes of the FileInfo follows the provisions indicated in table 4.2.1.4.29.2-1.

4.2.1.4.29.2 Attributes

Table 4.2.1.4.29.2-1 Attributes Properties for FileInfo

Attribute name	Data Type/Description	Properties
fileLocation	**Data Type:** string **Description:** Location of the file	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uri
fileSize	**Data Type:** integer **Description:** Size of the file, unit	x-support-qualifier: M readOnly: True x-isInvariant: False

Attribute name	Data Type/Description	Properties
	is byte	x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: int32 minimum: 0 x-units: bytes
fileReadyTime	**Data Type:** string **Description:** Date and time when the file was last closed and made available by the producer. The file content will not be changed any more	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: date-time
fileExpirationTime	**Data Type:** string **Description:** Date and time after which the file may be deleted.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: date-time
fileCompression	**Data Type:** string **Description**: If compression is applied to the file, the name of the compression algorithm used for compressing the file.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: True
fileDataType	**Data Type:** string **Description:** An enumerated set of values identifying the type of data contained in the file. The valid set of values are described as follows: PERFORMANCE: The file contains a PerformanceMeasurementReport. ANALYTIC: The file contains analytic data. LOG: The file contains log entries identified in a logQuery request.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False enum: - PERFORMANCE - ANALYTIC - LOG

4.2.1.4.29.3 Attribute constraints

None.

4.2.1.4.29.4 State diagram

None.

4.2.1.4.30 Location

4.2.1.4.30.1 Definition

The Location represents an instance of a Location where an O-Cloud Site may be available.

4.2.1.4.30.2 Attributes

Table 4.2.1.4.30.2-1 Attributes Properties for Location

Attribute name	Data Type/Description	Properties
globalLocationId	**Data Type:** string **Description:** Identifier of the location as defined by the SMO.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
name	**Data Type:** string **Description:** Human readable name of the location as defined by the SMO.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Human readable description of the location as defined by the SMO.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
oCloudId	**Data Type:** string **Description:** Reference to the O-Cloud available at the location.	support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
oCloudSiteIds	**Data Type:** array **Description:** List of O-Cloud Site identifiers referencing the O-Cloud Sites available at the location.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string
coordinate	**Data Type:** string **Description:** The latitude and longitude of the location (shall be supplied if a postal address not provided).	support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: True
address	**Data Type:** string **Description:** Postal address of the facility (shall be supplied if a coordinate is not provided).	support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: True
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the location.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeVa

Attribute name	Data Type/Description	Properties
		luePair

4.2.1.4.30.3 Attribute constraints

None.

4.2.1.4.30.4 State diagram

None.

4.2.1.4.31 OCloudSite

4.2.1.4.31.1 Definition

The OCloudSite represents an O-Cloud Site instance in an O-Cloud, as described in the O-RAN Cloud Architecture and Deployment Scenarios specification [i.3].

4.2.1.4.31.2 Attributes

Table 4.2.1.4.31.2-1 Attributes Properties for OCloudSite

Attribute name	Data Type/Description	Properties
oCloudSiteId	**Data Type:** string **Description:** Identifier of the O-Cloud site. Locally unique within the scope of an O-Cloud instance.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
locationId	**Data Type:** string **Description:** Reference to location where the O-Cloud site is deployed at.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
name	**Data Type:** string **Description:** Human readable name of the O-Cloud site as identified by the cloud provider.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Human readable description of the O-Cloud site as provided by the cloud provider.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
oCloudId	**Data Type:** string **Description:** Reference to the O-Cloud of which the O-Cloud site is part.	support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
resourcePools	**Data Type:** array	x-support-qualifier: M

Attribute name	Data Type/Description	Properties
	Description: List of resource pools that are part of the O-Cloud site.	readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: ResourcePool
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the O-Cloud site.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.1.4.31.3 Attribute constraints

None.

4.2.1.4.31.4 State diagram

None.

4.2.1.4.32 LogQuery

4.2.1.4.32.1 Definition

The LogQuery represents a request for O-Cloud log files for specific resources or resource of specific ResourceTypes to be collected and made available to the requestor.

4.2.1.4.32.2 Attributes

Table 4.2.1.4.32.2-1 Attributes Properties for LogQuery

Attribute name	Data Type/Description	Properties
logType	**Data Type:** string **Description:** This identifies the type of log(s) to be collected. The exact list of valid log types to be discovered via a log capability discovery.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
dateRanges	**Data Type:** array **Description:** list of date ranges that log data is to be extracted and placed into the log file for collection.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 items: DateRange
eventTypes	**Data Type:** array	x-support-qualifier: M

Attribute name	Data Type/Description	Properties

	Description: List of event types that are to be extracted and placed in the log file for collection. The exact list of valid event types are to be discovered via a log capability discovery.	`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: string`

resources	**Data Type:** array
	
	Description: list of resource selection criteria to identify which resources are in scope for the query.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: AttributeValuePair`

resourceTypes	**Data Type:** array
	
	Description: list of resourceType selection criteria to identify which resources matching the resourceTypes are in scope for the query.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: AttributeValuePair`

logLevels	**Data Type:** array
	
	Description: List of logging levels that are to be extracted and placed in the log file for collection. The log levels specified in Clause 6.2.1, Table 2 of IETF RFC 5424 [5] shall be supported by the O-Cloud instance.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: string`

remoteFileLocation	**Data Type:** string
	
	Description: When the service is expected to push to a remote file system, this field is populated with the URL to the remote directory, otherwise the field is empty or not present.
		`support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`format: uri`

callback	**Data Type:** string
	
	Description: The URL of the callback to which asynchronous notifications are delivered to.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`format: uri`

consumerLogRequestId	**Data Type:** string
	
	Description: Identifier optionally provided by the consumer for its purpose of tracking a request for log data.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

4.2.1.4.32.3 Attribute constraints

None.

4.2.1.4.32.4 State diagram

None.

4.2.1.4.33 DateRange <<dataType>>

4.2.1.4.33.1 Definition

The DateRange represent a start and end date and time value to indicate a span of time.

4.2.1.4.33.2 Attributes

Table 4.2.1.4.33.2-1 Attributes Properties for DateRange

Attribute name	Data Type/Description	Properties
startDateTime	**Data Type:** string **Description:** This identifies the start of a time range.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: date-time
endDateTime	**Data Type:** string **Description:** This identifies the end of a time range.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: date-time

4.2.1.4.33.3 Attribute constraints

Table 4.2.1.4.33.3-2 Attributes Constraints for DateRange

Name	Definition
START_PRECEDES_END	Condition: startDateTime shall chronologically precede the endDateTime.

.

4.2.1.4.33.4 State diagram

None.

4.2.2 Namespace: ORAN.O2ims.Artifacts

4.2.2.1 Namespace Overview

The O-RAN O2ims Artifacts namespace defines the classes and their attributes which are directly exposed to the consumer (SMO) via its O2 IMS Interface.

This namespace is used for the O-Cloud artifacts e.g. Cluster Templates, Initial Configuration Sets, OS/SW/FW, etc. that e.g. The O-Cloud artifacts can be used for configuring O-Cloud Resources from the Inventory and Cluster namespaces. The artifacts are not consumed when they are used and can be referenced in the Inventory, Cluster and Provisioning structures.

4.2.2.2 Imported and associated information entities

4.2.2.2.1 Imported information entities and local labels

Information entities imported into the model are also elements of the namespace and therefore shall remain unique within the namespace.

Table 4.2.2.2.1-1 Imported Entities

Label reference	Local label

4.2.2.2.2 Associated information entities and local labels

Information entities referenced in the model through an association which may be extended and overloaded with a new entity of the same name within the namespace are listed in the associated entities table.

Table 4.2.2.2.2-1 Associated Entities

Label reference	Local label
4.2.1.2.1 `Top`	Top
4.2.1.4.7, InfrastructureInventoryObject_	InfrastructureInventoryObject_

4.2.2.3 Class Diagram

4.2.2.3.1 Relationships

4.2.2.3.1.1 Description

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for the O-RAN Cloud (O-Cloud). This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes. The UML Class diagram depicts a view of the namespace maintaining a defined focus so as not to overcrowd a single diagram trying to depict all relationships within the namespace.

4.2.2.3.1.2 ORAN.O2ims.Artifacts Intra-Namespace Relationships

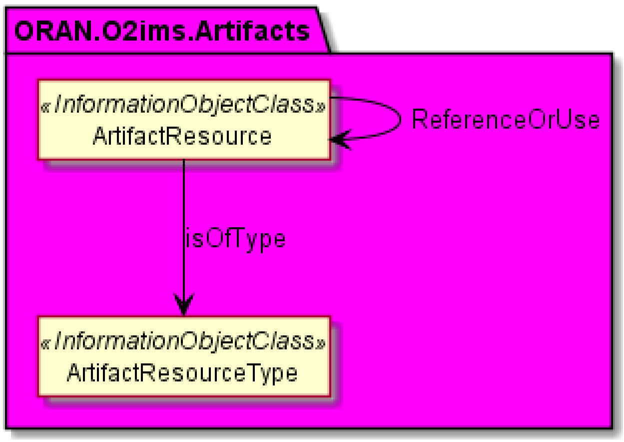

Figure 4.2.2.3.1.2-1 ORAN.O2ims.Artifacts Intra-Namespace Relationships

4.2.2.3.1.3 ORAN.O2ims.Artifacts Inter-Namespace Relationships

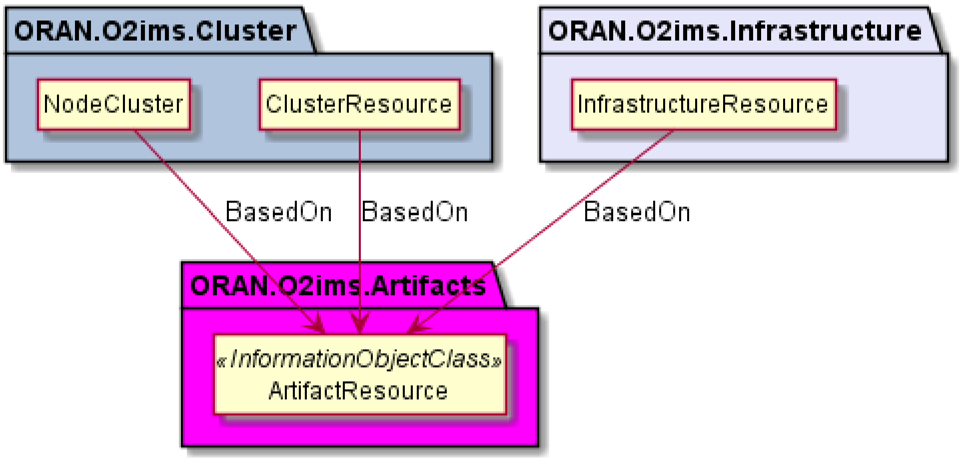

Figure 4.2.2.3.1.3-1 ORAN.O2ims.Artifacts Inter-Namespace Relationships

4.2.2.3.2 Inheritance

4.2.2.3.2.1 ORAN.O2ims.Artifacts Inheritance

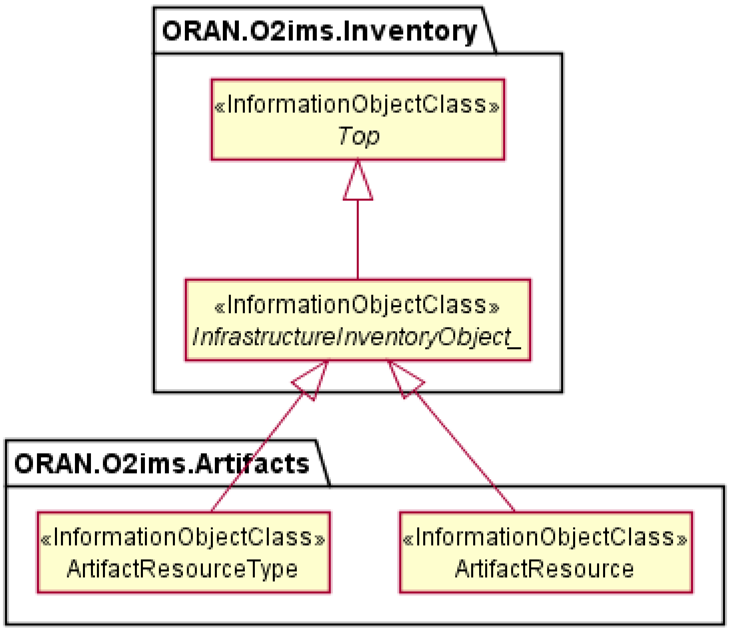

Figure 4.2.2.3.2.1-1 ORAN.O2ims.Artifacts Inheritance

4.2.2.4 Class Definitions

4.2.2.4.1 ArtifactResourceType

4.2.2.4.1.1 Definition

The ArtifactResourceType represents a resource type for a stored artifact resource in an O-Cloud.

Artifact resource types provide the categorization of O-Cloud artifacts used as declarative templates or binary images for provisioning or deployment. The values for these types are used to build the inventory with provisioning enablers. The O-Cloud provides a mechanism to discover data about artifacts with respective operations on artifact resource types.

The ArtifactResourceType is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The ArtifactResourceType inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.2.4.1.2 Attributes

Table 4.2.2.4.1.2-1 Attributes Properties for ArtifactResourceType

Attribute name	Data Type/Description	Properties

artifactResourceTypeId	**Data Type:** string
	
	Description: Cloud unique identifier for the artifact type.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Human readable name for the artifact type.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Description of the artifact type.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: False`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the artifact resource type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: AttributeValuePair`

4.2.2.4.1.3 Attribute constraints

None.

4.2.2.4.1.4 State diagram

None.

4.2.2.4.2 ArtifactResource

4.2.2.4.2.1 Definition

The ArtifactResource represents a stored artifact resource in an O-Cloud.

ArtifactResource provides the ability to expose information about the artifact that is needed by consumers without the need of parsing the artifact itself by the consumer. The O-Cloud provides a mechanism to discover new artifacts with respective operations on artifact resources.

The ArtifactResource is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The ArtifactResource inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.2.4.2.2 Attributes

Table 4.2.2.4.2.2-1 Attributes Properties for ArtifactResource

Attribute name	Data Type/Description	Properties
artifactResourceId	**Data Type:** string **Description:** O-Cloud unique identifier for the artifact.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
name	**Data Type:** string **Description:** Human readable name of the artifact.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
version	**Data Type:** string **Description:** Version of the artifact.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Description of the artifact.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
parameterSchema	**Data Type:** array **Description:** Properties (keys) which are tailored by the artifact composition when it is to be used as a parameterized template. See NOTE.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair
NOTE: Since the ArtifactResource is generic for any kind of artifact at an information model level, no keys are specified since this would have dependency on the kind of resource to which the artifact applies.		

4.2.2.4.2.3 Attribute constraints

None.

4.2.2.4.2.4 State diagram

None.

4.2.3 Namespace: ORAN.O2ims.Cluster

4.2.3.1 Namespace Overview

The O-RAN O2ims Cluster namespace defines the classes and their attributes which are directly exposed to the consumer (SMO) via its O2 IMS Interface.

This namespace is used for the provisioned O-Cloud Node Clusters with status and metrics as well as its internal topologies. The O-Coud Node Clusters are assembled based on the ORAN.O2ims.Inventory objects and configured through ORAN.O2ims.Artifacts.

4.2.3.2 Imported and associated information entities

4.2.3.2.1 Imported information entities and local labels

Information entities imported into the model are also elements of the namespace and therefore shall remain unique within the namespace.

Table 4.2.3.2.1-1 Imported Entities

Label reference	Local label

4.2.3.2.2 Associated information entities and local labels

Information entities referenced in the model through an association which may be extended and overloaded with a new entity of the same name within the namespace are listed in the associated entities table.

Table 4.2.3.2.2-1 Associated Entities

Label reference	Local label
4.2.1.4.7 *InfrastructureInventoryObject_*	*InfrastructureInventoryObject_*

4.2.3.3 Class Diagram

4.2.3.3.1 Relationships

4.2.3.3.1.1 Description

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for the O-RAN Cloud (O-Cloud). This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes. The UML Class diagram depicts a view of the namespace maintaining a defined focus so as not to overcrowd a single diagram trying to depict all relationships within the namespace.

4.2.3.3.1.2 ORAN.O2ims.Cluster Intra-Namespace Relationships

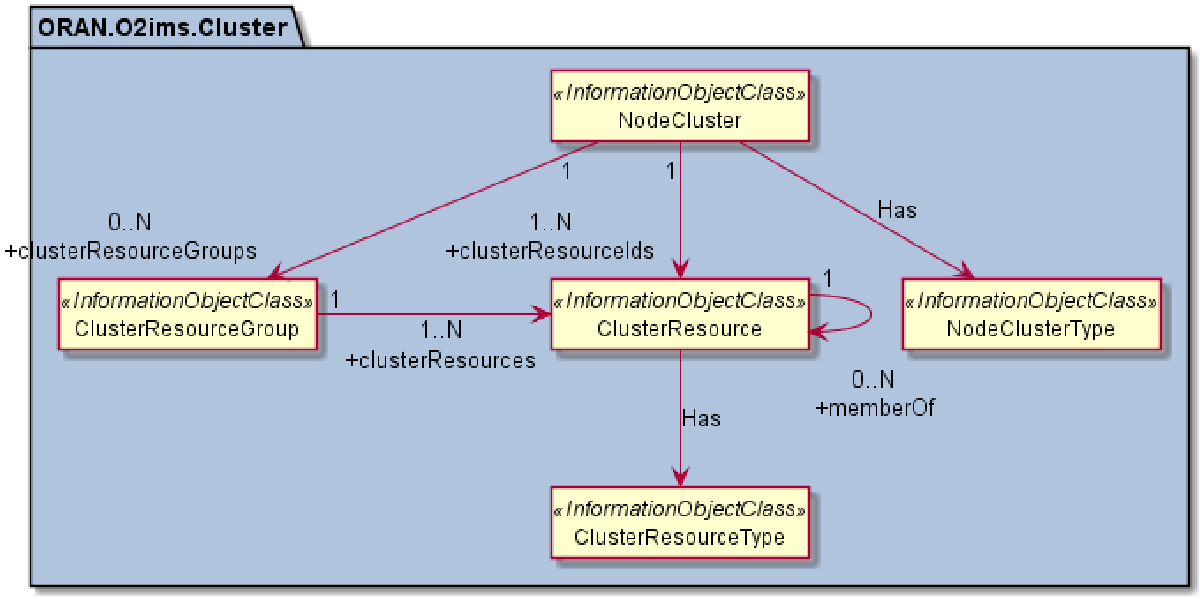

Figure 4.2.3.3.1.2-1 ORAN.O2ims.Cluster Intra-Namespace Relationships

4.2.3.3.1.3 ORAN.O2ims.Cluster Inter-Namespace Relationships

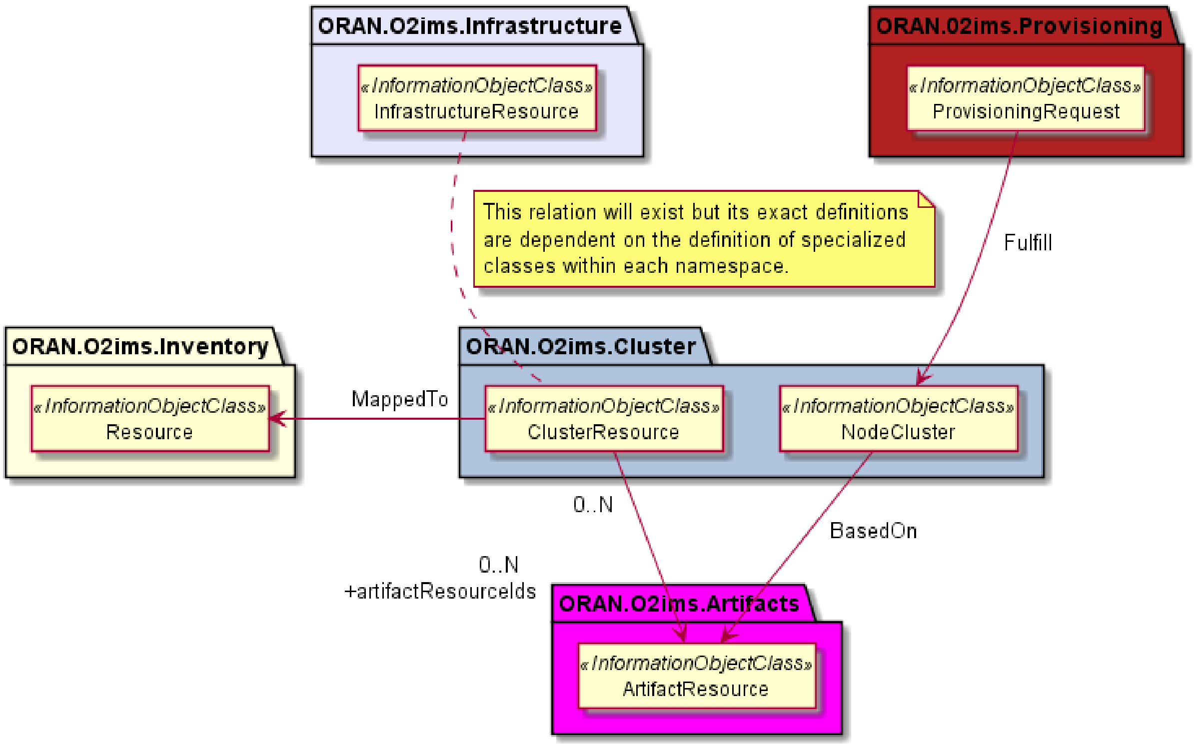

Figure 4.2.3.3.1.3-1 ORAN.O2ims.Cluster Inter-Namespace Relationships

4.2.3.3.2 Inheritance

4.2.3.3.2.1 ORAN.O2ims.Cluster Inheritance

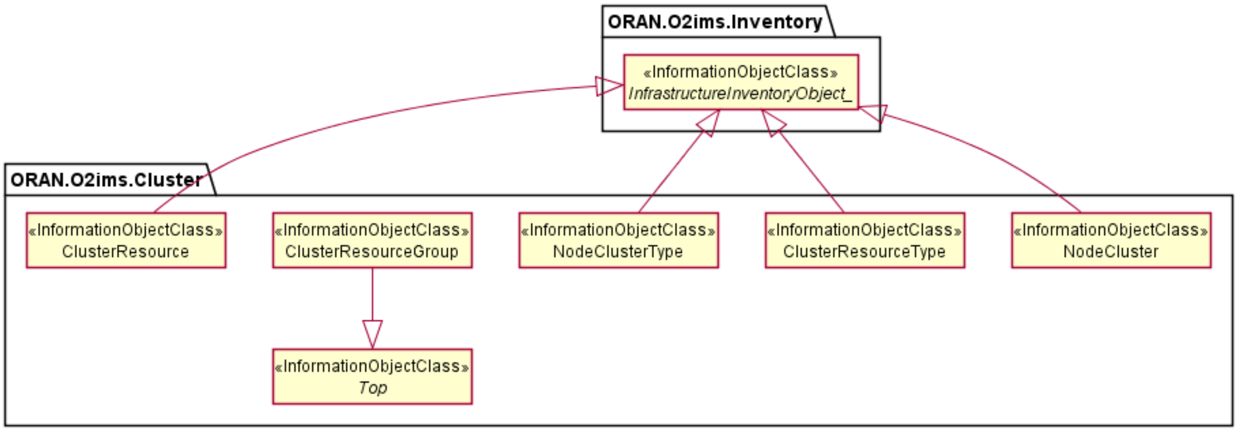

Figure 4.3.2.2.2.1-1 ORAN.O2ims.Cluster Inheritance

4.2.3.4 Class Definitions

4.2.3.4.1 NodeClusterType

4.2.3.4.1.1 Definition

The NodeClusterType represents a resource type for a NodeCluster in an O-Cloud.

Node cluster types provide the categorization of O-Cloud resource types used to create a node cluster. The O-Cloud provides a mechanism to discover the node cluster types with respective operations on node cluster types.

The NodeClusterType is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The NodeClusterType inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.3.4.1.2 Attributes

Table 4.2.3.4.1.2-1 Attributes Properties for NodeClusterType

Attribute name	Data Type/Description	Properties

NodeClusterTypeId	**Data Type:** string
	
	Description: Unique identifier for the NodeClusterType instance.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Name of the Node Cluster Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the Node Cluster Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

Attribute name	Data Type/Description	Properties

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored extend the information provided about the Node Cluster Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items:`
		`AttributeValuePair`

4.2.3.4.1.3 Attribute constraints

None.

4.2.3.4.1.4 State diagram

None.

4.2.3.4.2 NodeCluster

4.2.3.4.2.1 Definition

The NodeCluster represents a set if resource used to deploy workloads in an O-Cloud.

Node cluster provides inventory of the resources used in a NodeCluster. The O-Cloud provides a mechanism to discover the node cluster with respective operations on node clusters.

The NodeCluster is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The NodeCluster inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.3.4.2.2 Attributes

Table 4.2.3.4.2.2-1 Attributes Properties for NodeCluster

Attribute name	Data Type/Description	Properties

nodeClusterId	**Data Type:** string
	
	Description: Unique identifier for the NodeCluster instance assigned by the O-Cloud.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

clientNodeClusterId	**Data Type:** string
	
	Description: Unique identifier for the NodeCluster instance assigned by the consumer who requested its creation.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

name	**Data Type:** string
	
	Description: Name of the NodeCluster.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string **Description:** Human readable	`x-support-qualifier: M` `x-isInvariant: False`

Attribute name	Data Type/Description	Properties
	description of the NodeCluster.	x-inventoryNotification: True x-stateChangeNotification: False nullable: False
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored extend the information provided about the NodeCluster.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair
clusterDistributionDescription	**Data Type:** string **Description:** Human readable text identifying the method of distribution of ClusterResources over O-Cloud Sites.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
NodeClusterTypeId	**Data Type:** string **Description:** Identifier for the NodeClusterType that this NodeCluster is.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
artifactResourceId	**Data Type:** string **Description:** Identifier for the template artifact which this NodeCluster is based on.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
clusterResourceIds	**Data Type:** array **Description:** The list of cluster resource identifiers that are used to construct this NodeCluster.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: string format: uuid
clusterResourceGroups	**Data Type:** array **Description:** Optional list node groups that comprise the cluster resources which compose the NodeCluster.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string format: uuid

4.2.3.4.2.3 Attribute constraints

None.

4.2.3.4.2.4 State diagram

None.

4.2.3.4.3 ClusterResourceType

4.2.3.4.3.1 Definition

The ClusterResourceType represents a resource type for a ClusterResource in an O-Cloud.

Cluster resource types provide the categorization of O-Cloud resource types used to create a node cluster. The O-Cloud provides a mechanism to discover the cluster resource types with respective operations on cluster resource types.

The ClusterResourceType is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The ClusterResourceType inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.3.4.3.2 Attributes

Table 4.2.3.4.3.2-1 Attributes Properties for ClusterResourceType

Attribute name	Data Type/Description	Properties

clusterResourceTypeId	**Data Type:** string
	
	Description: Unique identifier for the ClusterResourceType instance.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Name of the Cluster Resource Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the Cluster Resource Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored extend the information provided about the Cluster Resource Type.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: AttributeValuePair`

4.2.3.4.3.3 Attribute constraints

None.

4.2.3.4.3.4 State diagram

None.

4.2.3.4.4 ClusterResource

4.2.3.4.4.1 Definition

The ClusterResource represents a resource used to construct a NodeCluster in an O-Cloud.

Node cluster resource provides inventory of the resources used in a NodeCluster. The O-Cloud provides a mechanism to discover the node cluster resources with respective operations on node cluster resources.

The ClusterResource is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The ClusterResource inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.3.4.4.2 Attributes

Table 4.2.3.4.4.2-1 Attributes Properties for ClusterResource

Attribute name	Data Type/Description	Properties

clusterResourceId	**Data Type:** string
	
	Description: Unique identifier for the ClusterResource instance.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Name of the cluster resource.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the cluster resource.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized) properties (keys) which are tailored extend the information provided about the Cluster Resource.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: AttributeValuePair`

clusterResourceTypeId	**Data Type:** string
	
	Description: Identifier for the ClusterResourceType for this resource.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

memberOf	**Data Type:** array
	
	Description: list of other ClusterResources which are linked together.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: string`

Attribute name	Data Type/Description	Properties

artifactResourceIds	**Data Type:** array
	
	Description: Identifiers for the artifact(s) which this resource is based on.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: string`
		`format: uuid`

resourceId	**Data Type:** string
	
	Description: Identifier for the inventory resource which this resource is mapped to.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

4.2.3.4.4.3 Attribute constraints

None.

4.2.3.4.4.4 State diagram

None.

4.2.3.4.5 ClusterResourceGroup

4.2.3.4.5.1 Definition

The ClusterResourceGroup represents a collection of ClusterResources as a named set.

4.2.3.4.5.2 Attributes

Table 4.2.3.4.5.2-1 Attributes Properties for ClusterResourceGroup

Attribute name	Data Type/Description	Properties

clusterResourceGroupId	**Data Type:** string
	
	Description: Unique identifier for the ClusterResourceGroup.
		`x-support-qualifier: M`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`
		`format: uuid`

name	**Data Type:** string
	
	Description: Name of the Node Group.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

description	**Data Type:** string
	
	Description: Human readable description of the ClusterResourceGroup.
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: False`

extensions	**Data Type:** array
	
	Description: These are unspecified (not standardized)
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`

Attribute name	Data Type/Description	Properties

	properties (keys) which are tailored extend the information provided about the ClusterResourceGroup.	`nullable: True`
		`x-isOrdered: False`
		`minItems: 0`
		`uniqueItems: True`
		`items: AttributeValuePair`

clusterResources	**Data Type:** array
	
	Description: The list of Cluster Resources belonging to this ClusterResourceGroup.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`x-isOrdered: False`
		`minItems: 1`
		`uniqueItems: True`
		`items: string`
		`format: uuid`

4.2.3.4.5.3 Attribute constraints

None.

4.2.3.4.5.4 State diagram

None.

4.2.4 Namespace: ORAN.O2ims.Provisioning

4.2.4.1 Namespace Overview

The O-RAN O2ims Provisioning namespace defines the classes and their attributes which are directly exposed to the consumer (SMO) via its O2 IMS Interface.

This namespace is used for provisioning requests and fulfillments, e.g. of O-Cloud Node Clusters, representing the requested and agreed service levels and their current identifiers.

4.2.4.2 Imported and associated information entities

4.2.4.2.1 Imported information entities and local labels

Information entities imported into the model are also elements of the namespace and therefore shall remain unique within the namespace.

Table 4.2.4.2.1-1 Imported Entities

Label reference	Local label
TS 28.622 [6], information object class, `Top`	`Top`

4.2.4.2.2 Associated information entities and local labels

Information entities referenced in the model through an association which may be extended and overloaded with a new entity of the same name within the namespace are listed in the associated entities table.

Table 4.2.4.2.2-1 Associated Entities

Label reference	Local label

4.2.4.3 Class Diagram

4.2.4.3.1 Relationships

4.2.4.3.1.1 Description

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for the O-RAN Cloud (O-Cloud). This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes. The UML Class diagram depicts a view of the namespace maintaining a defined focus so as not to overcrowd a single diagram trying to depict all relationships within the namespace.

4.2.4.3.1.2 ORAN.O2ims.Provisioning Intra-Namespace Relationships

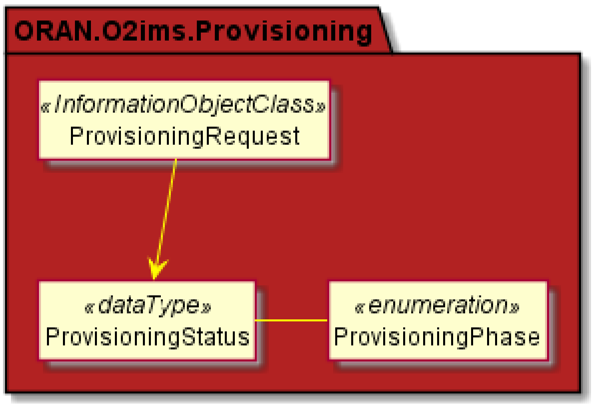

Figure 4.2.4.3.1.2-1 ORAN.O2ims.Provisioning Intra-Namespace Relationships

4.2.4.3.1.3 ORAN.O2ims.Provisioning Inter-Namespace Relationships

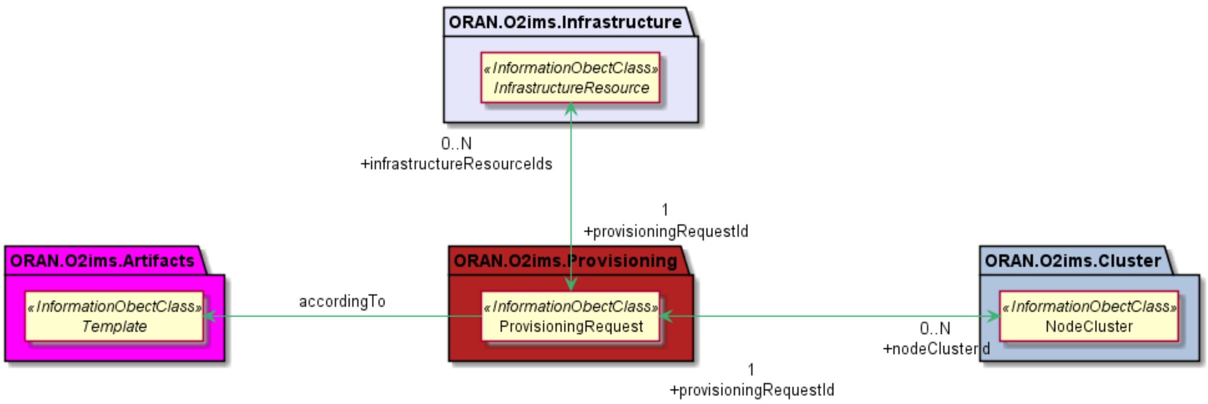

Figure 4.2.4.3.1.3-1 ORAN.O2ims.Provisioning Inter-Namespace Relationships

4.2.4.3.2 Inheritance

4.2.4.3.2.1 ORAN.O2ims.Provisioning Inheritance

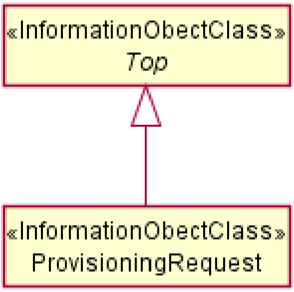

Figure 4.2.4.3.2.1-1 ORAN.O2ims.Provisioning Inheritance

4.2.4.4 Class Definitions

4.2.4.4.1 ProvisioningRequest

4.2.4.4.1.1 Definition

The ProvisioningRequest represents a request for item(s) to be provisioned in the O-Cloud. The SMO makes the declarative request to the O-Cloud through the O2ims.

4.2.4.4.1.2 Attributes

Table 4.2.4.4.1.2-1 Attributes Properties for ProvisioningRequest

Attribute name	Data Type/Description	Properties
provisioningRequestId	**Data Type:** string **Description:** Identifier of the provisioning request instance assigned by the SMO.	x-support-qualifier: M x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False format: uuid
name	**Data Type:** string **Description:** Human readable name of the provisioning request instance as assigned by the SMO.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Human readable description of the provisioning request instance as assigned by the SMO.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
templateName	**Data Type:** string **Description:** Name of the template to be used as the basis for the declarative model for the provisioning request instance. When combined with the templateVersion it shall resolve to a templateId.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
templateVersion	**Data Type:** string	x-support-qualifier: M

Attribute name	Data Type/Description	Properties
	Description: Version of the template used as the basis for the declarative model for the provisioning request instance. When combined with the templateName it shall resolve to a templateId.	x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
templateParameters	**Data Type:** array **Description:** List of variables/value pairs used within the template to tailor the template to this request instance.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair
provisioningRequest Reference	**Data Type:** string **Description:** A unique reference of this ProvisioningRequest assigned by the service producer at the time of request creation.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: False
status	**Data Type:** ProvisioningStatus **Description:** The time of the last message and the current provisioningPhase of the ProvisioningRequest.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False
provisionedResource Set	**Data Type:** provisionedResourceSet **Description:** The resources that have been successfully provisioned as part of the provisioningRequest.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True

4.2.4.4.1.3 Attribute constraints

Table 4.2.4.4.1.3-3 Attributes Constraints for ProvisioningRequest

Name	Definition
ReadOnlyAfterCreate	The following attributes are provided by the SMO on the create request but are readOnly aftwards: provisioningRequestId

4.2.4.4.1.4 State diagram

None.

4.2.4.4.2 ProvisioningStatus <<dataType>>

4.2.4.4.2.1 Definition

The ProvisioningStatus represents the status of a ProvisioningRequest indicating the time of the last message and the current provisioningPhase. The provisioningPhase is an enumerated set of values – PENDING, PROGRESSING, FULFILLED, FAILED, DELETING, that reflect the current phase of the provisioning request. The high-level definition of each provisioningPhase, are:

- PENDING: The ProvisioningRequest is waiting to be processed by the O-Cloud (IMS).
- PROGRESSING: The O-Cloud (IMS) is processing the ProvisioningRequest and executing the actions to fulfill it.
- FULFILLED: The ProvisioningRequest has been successfully processed and completed by the O-Cloud (IMS).
- FAILED: The ProvisioningRequest could not be fully processed by the O-Cloud (IMS).
- DELETING: The ProvisioningRequest is in the process of being deleted by the O-Cloud (IMS).

4.2.4.4.2.2 Attributes

Table 4.2.4.4.2.2-1 Attributes Properties for ProvisioningStatus

Attribute name	Data Type/Description	Properties
updateTime	**Data Type:** string **Description:** The last date and time that the provisioningPhase attribute was modified.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False format: date-time
message	**Data Type:** string **Description:** Human readable text about the provisioningPhase at the last updateTime.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False
provisioningPhase	**Data Type:** string **Description:** This is an enumerated set of values which reflects the current phase of the provisioning request.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False enum: - PENDING - PROGRESSING - FULFILLED - FAILED - DELETING

4.2.4.4.2.3 Attribute constraints

None.

4.2.4.4.2.4 State diagram

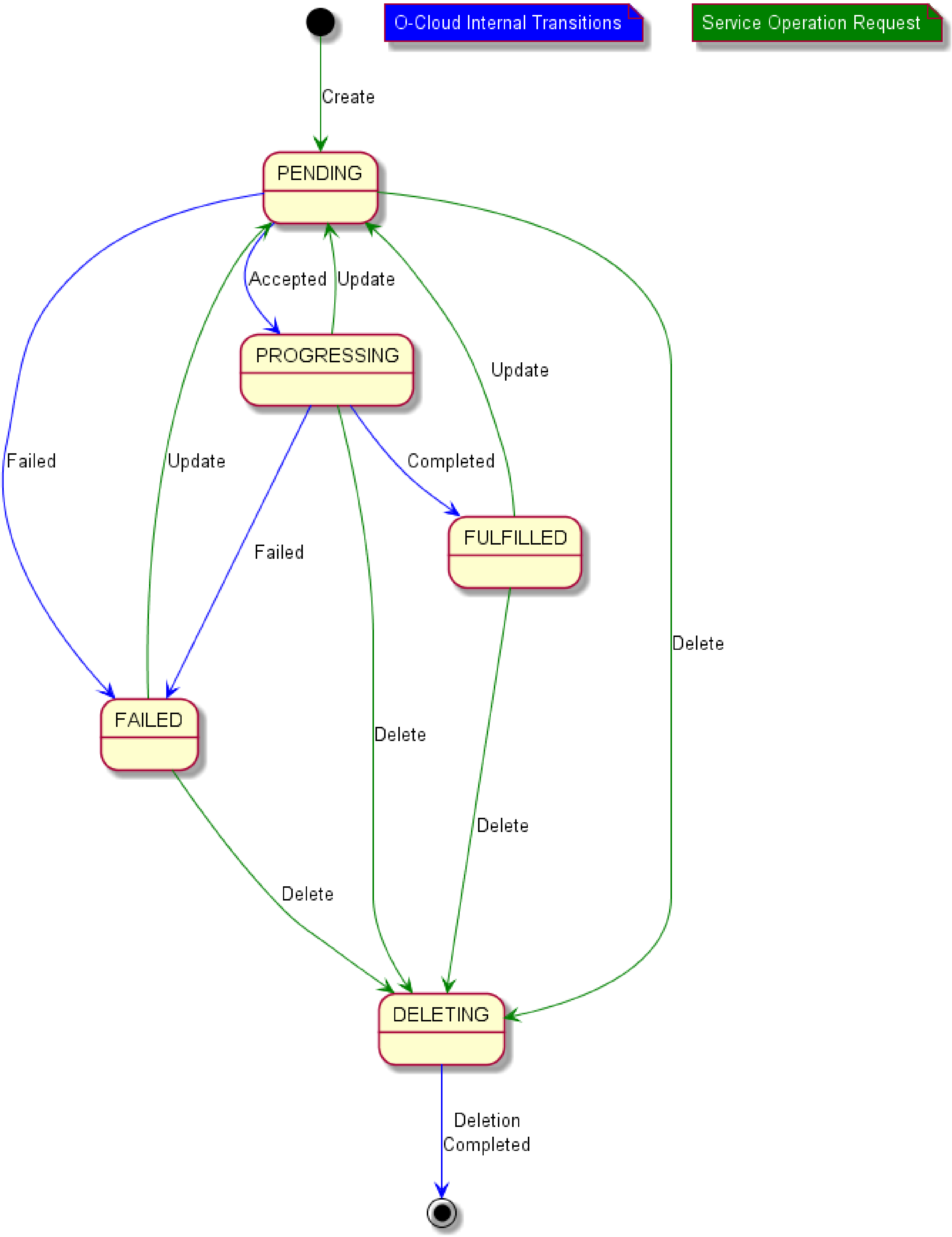

Figure 4.2.4.4.2.4-1 State Diagram for Provisioning Request (provisioningPhase) Phase Transition

4.2.4.4.3 ProvisionedResourceSet <<dataType>>

4.2.4.4.3.1 Definition

The ProvisionedResourceSet represents the resources that have been successfully provisioned as part of the ProvisioningRequest.

4.2.4.4.3.2 Attributes

Table 4.2.4.4.2.2-1 Attributes Properties for ProvisionedResourceSet

Attribute name	Data Type/Description	Properties
nodeClusterId	**Data Type:** String **Description:** If the request is fulfilled by a NodeCluster this field will contain its Id.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True format: uuid
infrastructureResourceIds	**Data Type:** array **Description:** If the request is fulfilled by InfrastructureResource(s) this list will contain the Id(s) of all resources used to fulfil it.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string

4.2.4.4.2.3 Attribute constraints

None.

4.2.4.4.2.4 State diagram

None.

4.2.5 Namespace: ORAN.O2ims.Infrastructure

4.2.5.1 Namespace Overview

The O-RAN O2ims Infrastructure namespace defines the classes and their attributes which are directly exposed to the consumer (SMO) via its O2 IMS Interface.

This namespace is used for fulfillment of provisioning requests which produce configured resources for later provisioning requests.

4.2.5.2 Imported and associated information entities

4.2.5.2.1 Imported information entities and local labels

Information entities imported into the model are also elements of the namespace and therefore shall remain unique within the namespace.

Table 4.2.5.2.1-1 Imported Entities

Label reference	Local label

4.2.5.2.2 Associated information entities and local labels

Information entities referenced in the model through an association which may be extended and overloaded with a new entity of the same name within the namespace are listed in the associated entities table.

Table 4.2.5.2.2-1 Associated Entities

Label reference	Local label
4.2.1.4.7, *InfrastructureInventoryObject_*	*InfrastructureInventoryObject_*

4.2.5.3 Class Diagram

4.2.5.3.1 Relationships

4.2.5.3.1.1 Description

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for the O-RAN Cloud (O-Cloud). This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes. The UML Class diagram depicts a view of the namespace maintaining a defined focus so as not to overcrowd a single diagram trying to depict all relationships within the namespace.

4.2.5.3.1.2 ORAN.O2ims.Infrastructure Intra-Namespace Relationships

The O-Cloud Networking consists of both intra- and inter-namespace relationships, which are complex in nature. The overview diagram (Figure 4.2.5.3.1.2-1) which simplifies the relationships has been drawn to help build a high-level understanding of the various object classes and their relationships. For example, a "Node" of type VM, BareMetal, and/or IaaS, which is a cluster resource type, terminates one or more site networks, which is an infrastructure resource type. Similarly, a gateway can terminate one or more site networks, forming an intra-namespace relationship since both the gateway and site networks belong to the infrastructure namespace.

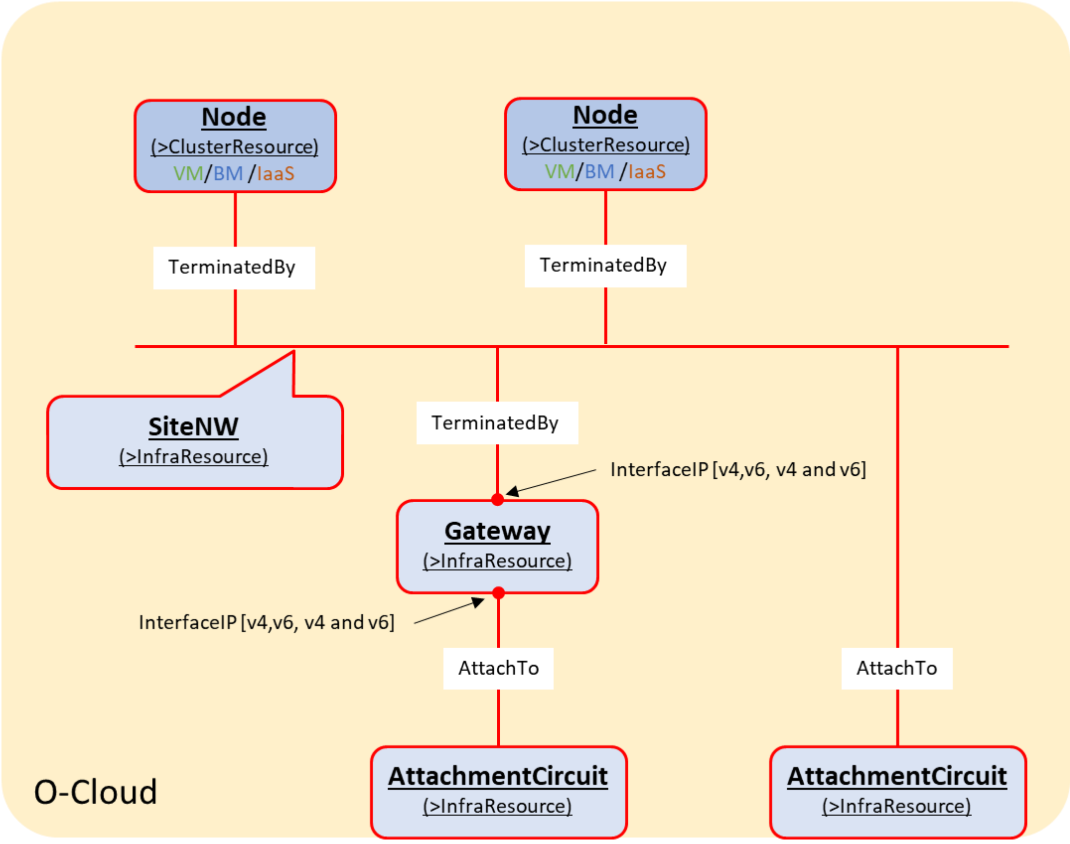

Figure 4.2.5.3.1.2-1 O-Cloud Networking Overview

These relationships, however, involve various elements and endpoints to support multiplicity, which introduces several implicit data types such as **Port**, **NetworkAddress**, and **SegmentationElement**. Each of these elements can include multiple endpoints representing a termination or an external connectivity attachment. These implicit data types and their relationships with O-Cloud Networking objects are captured in Figure 4.2.5.3.1.2-2, which illustrates the intra-namespace relationships between specialized infrastructure namespace objects.

There is a relationship between “Port” and “AttachmentCircuit” to represent the endpoint connectivity provided by “Port” for O-Cloud Site connectivity, using two possible associations to the transport network described in Clause 4.2.1.3.1.4.1.

Note: This document version represents the relationship with a dotted line, as the O-Cloud Networking Information Model is still in development.

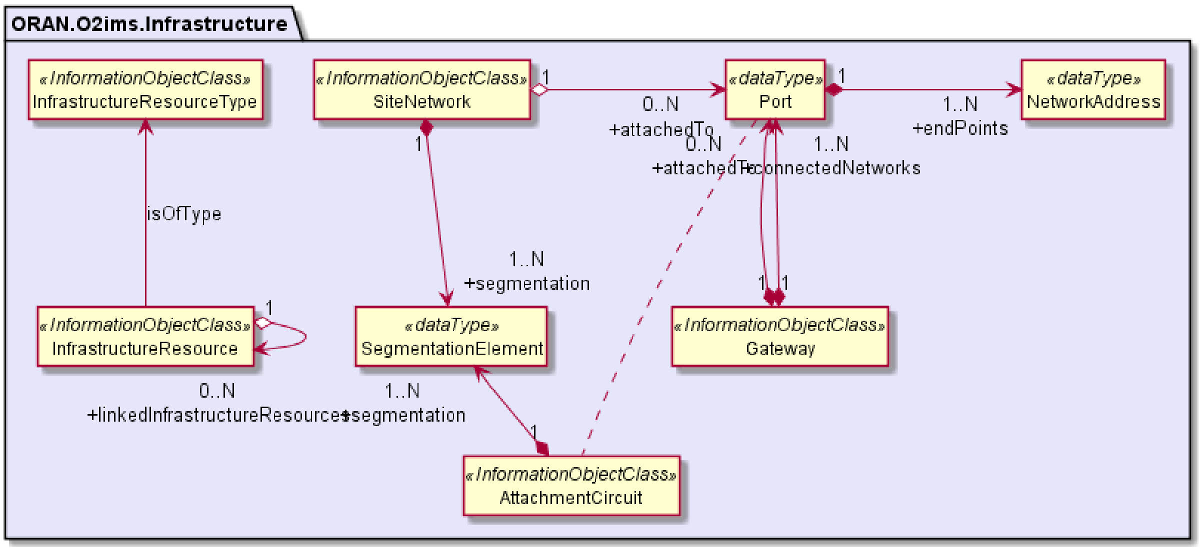

Figure 4.2.5.3.1.2-2 ORAN.O2ims.Infrastructure Intra-Namespace Relationships

4.2.5.3.1.3 ORAN.O2ims.Infrastructure Inter-Namespace Relationships

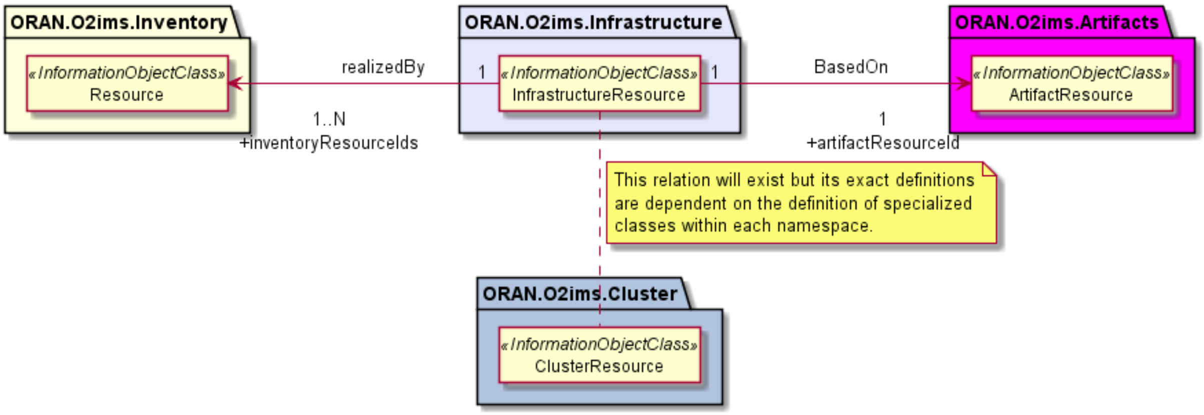

Figure 4.2.5.3.1.3-1 ORAN.O2ims.Infrastructure Inter-Namespace Relationships

4.2.5.3.2 Inheritance

4.2.5.3.2.1 ORAN.O2ims.Infrastructure Inheritance

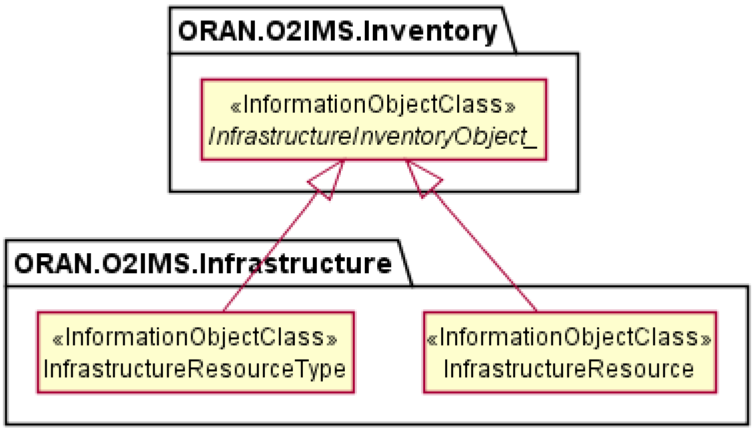

Figure 4.2.5.3.2.1-1 ORAN.O2ims.Infrastructure Inheritance

4.2.5.4 Class Definitions

4.2.5.4.1 InfrastructureResourceType

4.2.5.4.1.1 Definition

The InfrastructureResourceType represents a resource type in an O-Cloud.

Infrastructure resource types provide the categorization of O-Cloud resources used to connect the O-Cloud resources and to build components of the Cluster namespace. The values for these types are used to build the configured inventory and enable an authorized consumer to track infrastructure resource changes over time. The O-Cloud provides a mechanism to discover the infrastructure resource types with respective operations on infrastructure resource types.

The InfrastructureResourceType is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The InfrastructureResourceType inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.5.4.1.2 Attributes

Table 4.2.5.4.1.2-1 Attributes Properties for InfrastructureResourceType

Attribute name	Data Type/Description	Properties
infrastructureResourceTypeId	**Data Type:** string **Description:** Identifier of the infrastructure resource type.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
name	**Data Type:** string **Description:** Name of the infrastructure resource type as identified by the cloud provider.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Human readable	x-support-qualifier: M readOnly: True

Attribute name	Data Type/Description	Properties
	description of the infrastructure resource type.	x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Infrastructure Resource Type.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.5.4.1.3 Attribute constraints

None.

4.2.5.4.1.4 State diagram

None.

4.2.5.4.2 InfrastructureResource

4.2.5.4.2.1 Definition

The InfrastructureResource is a Specialized Object that derives from a Generalized Object, the *InfrastructureInventoryObject_*. The InfrastructureResource inherits the attributes from the *InfrastructureInventoryObject_* and allows it and all specializations of this class to be included in a change notification.

4.2.5.4.2.2 Attributes

Table 4.2.5.4.1.2-1 Attributes Properties for InfrastructureResource

Attribute name	Data Type/Description	Properties
infrastructureResourceId	**Data Type:** string **Description:** Identifier of the resource. Locally unique within the scope of an O-Cloud instance.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
name	**Data Type:** string **Description:** Human readable name of the infrastructure resource instance.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
description	**Data Type:** string **Description:** Human readable description of the infrastructure resource.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
infrastructureResourceTypeId	**Data Type:** string	x-support-qualifier: M

Attribute name	Data Type/Description	Properties
	Description: Identifier of infrastructure resource type that this instance is a type of.	readOnly: True x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
linkedInfrastructureResourceIds	**Data Type:** array **Description:** List of infrastructure resource IDs that are used to realize this infrastructure resource instance.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: string
inventoryResourceIds	**Data Type:** array **Description:** List of inventory resource IDs that are used to realize this infrastructure resource instance.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False x-isOrdered: False minItems: 1 uniqueItems: True items: string
artifactResourceId	**Data Type:** string **Description:** Identifier of artifact resource Id that the instance was configured with.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification: True x-stateChangeNotification: False nullable: False format: uuid
extensions	**Data Type:** array **Description:** These are unspecified (not standardized) properties (keys) which are tailored by the vendor to extend the information provided about the Resource Pool.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: True x-isOrdered: False minItems: 0 uniqueItems: True items: AttributeValuePair

4.2.5.4.2.3 Attribute constraints

None.

4.2.5.4.2.4 State diagram

None.

4.2.5.4.3 Gateway

4.2.5.4.3.1 Definition

The Gateway infrastructure resource is a specialization of the InfrastructureResource. As such in addition to the attributes below, all attributes of the InfrastructureResource are also a member of this class.

NOTE 1: The present document version does not specify QoS related attributes/classes.

NOTE 2: The present document version does not specify complete L2/L3 related attributes/classes associated to the Gateway.

The Gateway is a Specialized Object that derives from a Generalized Object, the InfrastructureResource. The Gateway inherits the attributes from the InfrastructureResource.

4.2.5.4.3.2 Attributes

Table 4.2.5.4.3.2-1 Attributes Properties for Gateway

Attribute name	Data Type/Description	Properties

tags	**Data Type:** array
	
	Description: The list of tags/labels on the Gateway resource.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: AttributeValuePair`

status	**Data Type:** string
	
	Description: The Gateway status. For instance, ACTIVE, DOWN, BUILD or ERROR.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: True`
		`nullable: False`

connectedNetworks	**Data Type:** array
	
	Description: The list of port(s) to support one or multiple Site Networks termination. A port is a list of Gateway interfaceIPs e.g. IPv4 and IPv6 for a Site Network.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: Port`

realizedBy	**Data Type:** string
	
	Description: Where this gateway is implemented/realized (e.g. SiteGW).
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: True`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`format: uuid`

attachedTo	**Data Type:** array
	
	Description: The list of port(s) to support one or multiple Attachment Circuits. A port is a list of Gateway interfaceIPs e.g. IPv4 and IPv6 for an Attachment Circuit.
		`x-support-qualifier: M`
		`readOnly: True`
		`x-isInvariant: False`
		`x-inventoryNotification: True`
		`x-stateChangeNotification: False`
		`nullable: True`
		`minItems: 0`
		`items: Port`

4.2.5.4.3.3 Attribute constraints

None.

4.2.5.4.3.4 State diagram

None.

4.2.5.4.4 SiteNetwork

4.2.5.4.4.1 Definition

The SiteNetwork infrastructure resource is a specialization of the InfrastructureResource. As such in addition to the attributes below, all attributes of the InfrastructureResource are also a member of this class.

NOTE: The present document version does not specify QoS related attributes/classes.

The SiteNetwork is a Specialized Object that derives from a Generalized Object, the InfrastructureResource. The SiteNetwork inherits the attributes from the InfrastructureResource.

4.2.5.4.4.2 Attributes

Table 4.2.5.4.4.2-1 Attributes Properties for SiteNetwork

Attribute name	Data Type/Description	Properties
tags	**Data Type:** array **Description:** The list of tags/labels on the SiteNetwork resource.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True items: AttributeValuePair
mtu	**Data Type:** integer **Description:** The maximum transmission unit (MTU) value to address fragmentation. Minimum value is 68 for IPv4, and 1280 for IPv6.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True
segmentation	**Data Type:** array **Description:** The list of segmentationElement(s) to support one or multiple segments where, a segmentationElement is a list of segmentationID (e.g. 100, 200) and segmentationType (e.g. flat, vlan, vxlan, or gre)	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False minItems: 1 items: SegmentationElement
realizedBy	**Data Type:** string **Description:** Where this network/segment is implemented/realized (e.g. SiteNWFabric).	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False
shared	**Data Type:** boolean **Description:** Indicates if the	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True

Attribute name	Data Type/Description	Properties
	network is shared (True).	x-stateChangeNotification: False nullable: False
status	**Data Type:** string **Description:** The SiteNetwork status. For instance, ACTIVE, DOWN, BUILD or ERROR.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: True nullable: False
subnets	**Data Type:** array **Description:** The list of associated subnets.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False minItems: 1 items: KeyValuePair
externalSiteNW	**Data Type:** boolean **Description:** Indicates if the network is also external (True) or not (False) to the O-Cloud Site.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True default: False
attachedTo	**Data Type:** array **Description:** The list of port(s) to support one or multiple Attachment Circuits. A port is a list of interfaceIPs e.g. IPv4 and IPv6 for an Attachment Circuit.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True minItems: 0 items: Port

4.2.5.4.4.3 Attribute constraints

None.

4.2.5.4.4.4 State diagram

None.

4.2.5.4.5 AttachmentCircuit

4.2.5.4.5.1 Definition

The AttachmentCircuit infrastructure resource is a specialization of the InfrastructureResource. As such in addition to the attributes below, all attributes of the InfrastructureResource are also a member of this class.

NOTE: The present document version does not specify QoS related attributes/classes.

The AttachmentCircuit is a Specialized Object that derives from a Generalized Object, the InfrastructureResource. The AttachmentCircuit inherits the attributes from the InfrastructureResource.

4.2.5.4.5.2 Attributes

Table 4.2.5.4.5.2-1 Attributes Properties for AttachmentCircuit

Attribute name	Data Type/Description	Properties
tags	**Data Type:** array **Description:** The list of tags/labels on the AttachmentCircuit resource.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: True minItems: 0 items: AttributeValuePair
segmentation	**Data Type:** array **Description:** The list of segmentationElement(s) to support one or multiple segments where, a segmentationElement is a list of segmentationID (e.g. 100, 200) and segmentationType (e.g. flat, vlan, vxlan, or gre)	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False minItems: 1 items: SegmentationElement
bearer	**Data Type:** string **Description:** A physical or logical link that establishes connectivity between the O-Cloud Site and external networks.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False

4.2.5.4.5.3 Attribute constraints

None.

4.2.5.4.5.4 State diagram

None.

4.2.5.4.6 Port <<dataType>>

4.2.5.4.6.1 Definition

The Port is a list of network addresses (IPv4, IPv6, or IPv4 and IPv6) which a network device is connected to.

4.2.5.4.6.2 Attributes

Table 4.2.5.4.6.2-1 Attributes Properties for Port

Attribute name	Data Type/Description	Properties
endPoints	**Data Type:** array **Description:** The list of addresses to support one or multiple network connections.	x-support-qualifier: M readOnly: True x-isInvariant: False x-inventoryNotification: True x-stateChangeNotification: False nullable: False minItems: 1

Attribute name	Data Type/Description	Properties
		items: NetworkAddress

4.2.5.4.6.3 Attribute constraints

None.

4.2.5.4.6.4 State diagram

None.

4.2.5.4.7 SegmentationElement <<dataType>>

4.2.5.4.7.1 Definition

The SegmentationElement represents one segment of a SiteNetwork. A SiteNetwork may have more than one segment.

4.2.5.4.7.2 Attributes

Table 4.2.5.4.7.2-1 Attributes Properties for SegmentationElement

Attribute name	Data Type/Description	Properties
segmentationType	**Data Type:** string **Description:** The description of the type of segmentation.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False
Segments	**Data Type:** array **Description:** list of segmentIds for a given SiteNetwork segment.	x-support-qualifier: M x-isInvariant: False x-inventoryNotification: False x-stateChangeNotification: False nullable: False minItems: 1 items: string

4.2.5.4.7.3 Attribute constraints

None.

4.2.5.4.7.4 State diagram

None.

4.2.5.4.8 NetworkAddress <<dataType>>

4.2.5.4.8.1 Definition

The NetworkAddress is an IP address (IPv4, IPv6, or IPv4 & IPv6) which identifies a network address.

4.2.5.4.8.2 Attributes

Table 4.2.5.4.8.2-1 Attributes Properties for NetworkAddress

Attribute name	Data Type/Description	Properties

ipv4Addr	**Data Type:** Ipv4Addr
	
	Description: IPv4 address as defined in 3GPP TS 28.623
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

ipv6Addr	**Data Type:** Ipv6Addr
	
	Description: IPv6 address as defined in 3GPP TS 28.623
		`x-support-qualifier: M`
		`x-isInvariant: False`
		`x-inventoryNotification: False`
		`x-stateChangeNotification: False`
		`nullable: True`

4.2.5.4.8.3 Attribute constraints

Table 4.2.5.4.8.3-1 Attributes Constraints for EndPoint

Name	Definition
ATLEAST_ONE_ADDRESS_EXISTS	Condition: if ipv4Addr exists then ipv6Addr is optional. Likewise, if ipv6Addr exists then ipv4Addr is optional.

4.2.5.4.8.4 State diagram

None.

Annex (informative): Change History

Date	Revision	Description
2023.03.21	00.01.00	Create initial O-RAN version, from template with modifications for 3GPP TS 32.160
2023.04.28	00.01.01	Modified Scope, Introduction, References, and Annex Formatinng CR: ATT-2023.03.15-WG6-CR-001-O2IM Introduction and Scope Changes - v02.
2023.04.28	00.01.02	Modified Section 4 Outline with examples CR: ATT-2023.04.04-CR-003-02IM Section 4 new Outline-v04.
2023.06.02	00.01.03	Incorporated the following CRs: ATT-2023.05.02-WG6-CR-004-ORIM Template Cleanup-v01 ATT-2023.05.03-WG6-CR-005-O2IM IMS Diagrams-v03 ATT-2023.05.23-WG6-CR-006-O2IM Imported and Associated Entities-v03 ATT-2023.05.23-WG6-CR-007-O2IM Fault Monitoring-v04
2023.07.10	00.02.00	Initial internal release based on the current O2IMS Specification. Incorporated the following CRs: ATT-2023.05.30-WG6-CR-008-O2IM Infrastructure Inventory-v05 ATT-2023.05.23-WG6-CR-009-O2IM LCM-v03 ATT-2023.05.31-WG6-CR-010-O2IM Performance Store-v03 ATT-2023.05.31-WG6-CR-011-O2IM Performance Job-v04 ATT-2023.06.01-WG6-CR-012-O2IM Performance Subscription-v04 ATT-2023.06.01-WG6-CR-013-O2IM Performance Report-v03
2024.03.12	01.00.00	Initial release Incorporated the following CRs: ATT-2024.03.01-WG6-CR-019-O2IM CR Merge Corrections-v03 DELL.AO-2023.09.26-WG6-CR-0002-O2IM_Site_v07 ATT-2024.01.26-WG6-CR-017-O2IM Namespace alignment-V02 ATT-2024.01.26-WG6-CR-018-O2IM Measurement Job Corrections-V04 ATT-2023.09.15-WG6-CR-016-O2IM Deferred Comment Corrections-V04 DELL.AO-2023.02.21-WG6-CR-0001-O2IM Extensions to infrastructure service objects-v03 ATT-2023.08.25-WG6-CR-015-O2IM New PM Use Case Support-v03 ATT-2023.06.01-WG6-CR-014-O2IM New Alarm Use Case Support-v06 Editorial, missing clauses for new section 4.3.3.5.3, 4.3.3.5.4
2024.07.01	01.00.04	Incorporated the following CRs: ERI.AO-2024.04.09-WG6-CR-0029-Namespace-Updates-v03 ATT-2024.04.15-WG6-CR-0020-New Namespace Skeletons-v03 ATT-2024.04.23-CR-021-Remove Performance Measurement Subscription-v01 ERI.AO-2023.07.25-WG6-CR-0024-OCloud_Networking_Information_Model-v21
2024.07.05	01.00.05	Incorporated the following CRs: ATT-2024.05.28-WG6-CR-0022-Performance Subscription Correction-v02 ATT-2024.05.28-WG6-CR-0023-log Query Enhancements-v05
2024.07.19	01.00.06	Incorporated Release review Comments
2024.07.30.	02.00.00	Release 2.0
2024.11.12	02.00.01	Incorporated the following CRs: DELL.AO-2024.08.27-WG6-CR-0010-O2IM_DictionaryId_v01 NOK.AO-2024.09.10-WG6-CR0131-GenSpclResource_Guidelines_v01.01 ATT-2024.07.30-WG6-CR-0024-Provisioning Relationship Modeling-v06 ATT.AO-2024.07.31-WG6-CR-0025-Networking Infrastructure Attribution-v09 ATT.AO-2024.09.11-WG6-CR-0026-Infrastructure Specialized Classes-v09 ATT-2024.09.11-WG6-CR-0027-PM Job Writeable Extensions-v02 ATT-2024.09.11-WG6-CR-0028-InfrastructureInventoryObject Name Correction-v03 ATT-2024.10.02-WG6-CR-0030-Cluster Classes and Attribution-v05 ATT-2024.10.02-WG6-CR-0031-Artifact Classes and Attribution-v04 ATT-2024.10.02-WG6-CR-0032-Provisioning Classes and Attribution-v08 ATT.AO-2024.10.15-WG6-CR-0033-3GPP Alignment to Release 18-v02 The following Editorial changes were made to make the document cohesive Boiler plate text required for the Infrastructure Namespace which was not included in ATT-CR-0025 or ATT-CR-0026. This included the clause 4.2.5 itself and clauses 4.2.5.1, 4.2.5.2, and 4.2.5.3
2024.11.26	03.00.00	Release 3.0

2025.03.11	03.00.01	Incorporated the following CR: DTAG-2025.01.24-WG6-CR-0002-O2IM_extensions attribute_v02.00 DELL.AO-2025.01.28-WG6-CR-0012-O2IM_Location_v03.00 ERI.AO-2025.01.13-WG6-CR-0056-O2IM-Update-ProvisioningRequest-InfoObjectClass-CR-v05
2025.03.18	03.00.02	Incorporated Release comments from Information Modeling Team. Many Font changes, especially to tables for consistency and 99lignment to the O-RAN template.
2025.03.21	03.00.03	Incorporated Release review Comments
2025.03.27	04.00.00	Release 4.0
2025.06.23	04.00.01	Incorporated the following CR: DTAG-29.04.2025-WG6-CR-0003_Update to NotifyFileReady and LogQuery_v02.00 DTAG.AO-23.05.2025-WG6-CR-0004 PM Job Change Notification_v01.00 DTAG.AO-23.05.2025-WG6-CR-0006 Consumer provided identifier format IM-v01.00 ERI.AO-2025.04.10-WG6-CR-0074-O2IM-ProvisioningRequest-InfoObjectClass-CR-v04
2025.06.24	04.00.02	Incorporated the following CR: RHT.2025.04.07-WG6-CR-0002-O2IM_Generic_Object_References_For_Alarms_v04.00
2025.07.07	04.00.03	Formatting changes to align with new O-RAN Template

